

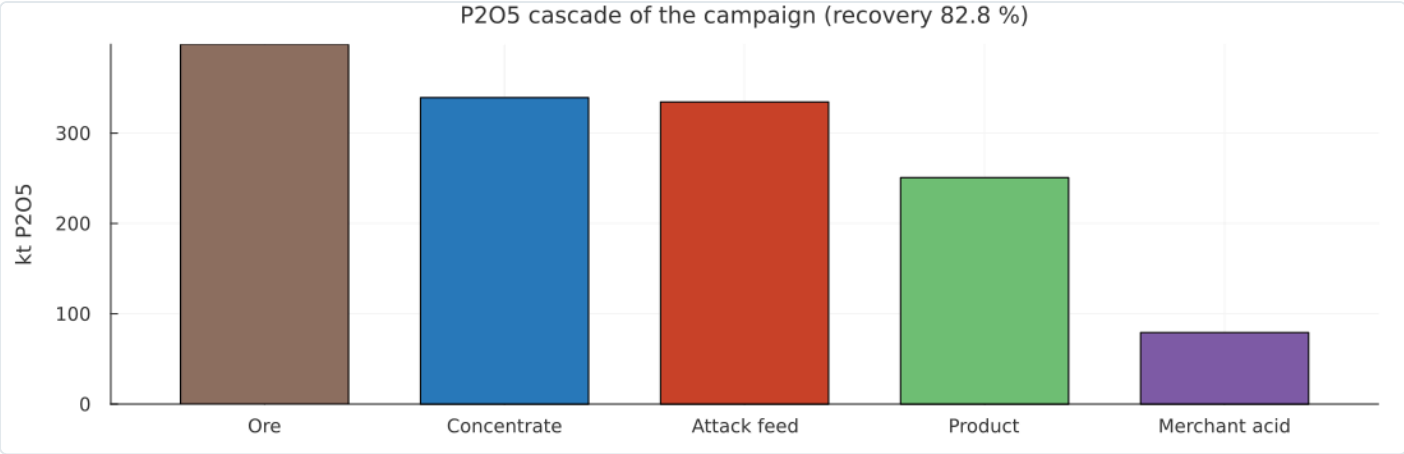
Atlas Phosphate Complex -- phosphate complex report

The ore, the acid, the product, the twin, the controller and the evidence

Site	Atlas Phosphate Complex (:atlas_phosphate)
Reporting period	2026-01-01T00:00:00 to 2026-04-30T23:00:00 (120 days, 2736 operating hours)
Product	Dap from 29.0 % P2O5 ore, 545.0 kt of product
Framework scope	ISO 50001:2018 energy management; ISO 14001:2015 environmental management; ISO 46001:2019 water efficiency; ISO 9001:2015 quality management; GHG Protocol corporate standard; EU Industrial Emissions Directive 2010/75; US EPA 40 CFR 63 subpart AA; IFA/EFMA BAT for phosphate fertilisers; Phosphogypsum stack management code
Provenance	Design revision A seed 20260101 2880 hourly intervals of 69 signals generated 2026-09-23T15:58:38.796 basis :screening_plant_model

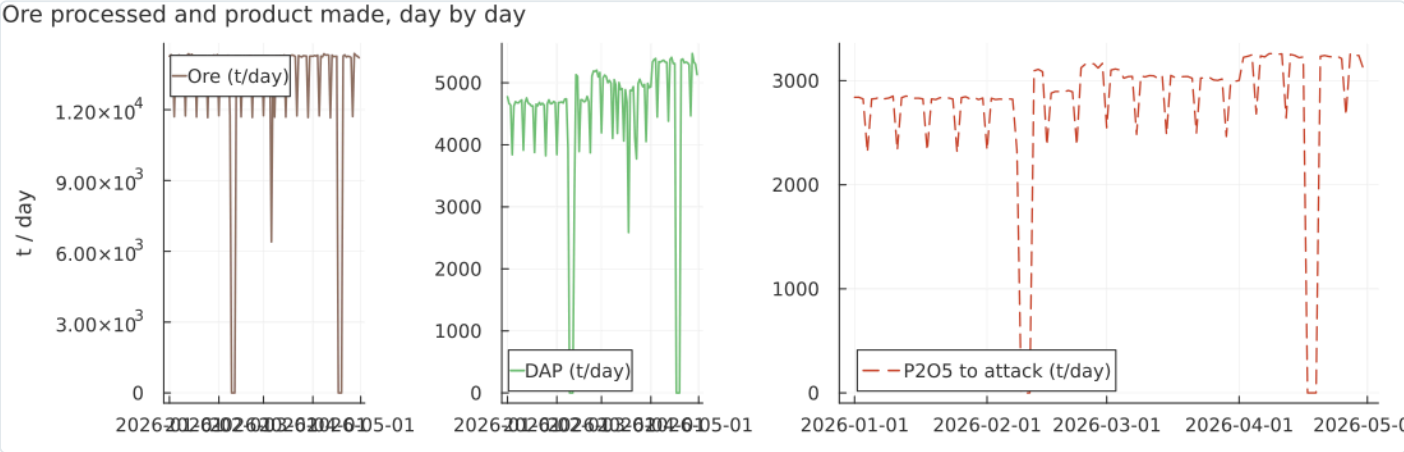
Executive summary

ORE PROCESSED 1578.582 kt 25.235 % P2O5 feed
CONCENTRATE 1067.699 kt 31.779 % P2O5, recovery 85.2 %
● P2O5 TO ATTACK 334.476 kt acid 2.847 t/t
● ACID STRENGTH 55.742 % P2O5 steam 0.408 t/t
● PRODUCT 544.501 kt 18.118 % N, 88.5 % WSP
● P2O5 RECOVERY 82.829 % plant, mine to bag
● SPECIFIC ENERGY 2.021 GJ/t P2O5 56.8 kWh/t
● VARIABLE COST 396.602 USD/t margin 651247.62 USD/t
● CARBON 55.923 kgCO2e/t scope 1+2, 19347.0 t total
● HISTORIAN 99.423 % usable 140 defect runs



The P2O5 cascade of the campaign, from the ore to the bagged product

Headline The complex processed **1,578.6 kt** of ore at 25.2 % P2O5 and delivered **544.5 kt** of Dap at 18.1 % N, recovering **82.8 %** of the P2O5 of the ore. The acid train digested 334.5 kt of P2O5 with 2.85 t of sulphuric acid per tonne, the filters lost 2.34 % of it in the gypsum and the evaporators spent 0.4075 t of steam per tonne. The tonne of product cost **396.6 :USD** in variable cost and carried 55.92 kgCO2e per tonne of P2O5. The detector raised 63 findings worth 7.7162261e7 :USD a year, and the plant is **Noncompliant** against 42 clauses of 9 frameworks.



Ore processed and product made, day by day, over the campaign

The plant: flowsheet, design and instrumentation

REGISTERED DESIGN

Quantity	Value
Site	Atlas Phosphate Complex (Morocco)
Product	Dap
Design capacity	1,200 kt P2O5 per year over 8,000 h
Design rate	150 t P2O5/h, 601.44 t/h of ore
Registered units	34 unit operations in 9 areas, 57 streams
Instrument register	69 tags, 9 of them around the attack tanks
Ore bodies	boucraa, florida_pebble, gafsa, jacobina, khouribga, tapira
Revision	A -- capacities and targets are representative screening values

Unit	Area	Equipment	Kind	Feeds	Products	Capacity
:CRUSHING	:beneficiation	:jaw_crusher	:crush	:rom_ore	:crushed_rock	448
:GRINDING	:beneficiation	:rod_mill	:grind	:crushed_rock, :dilution_water	:mill_discharge	320
:CYCLONE_PACK	:beneficiation	:hydrocyclone_pack	:classify	:mill_discharge	:cyclone_overflow, :cyclone_underflow	320
:DESLIMING	:beneficiation	:desliming_cyclone	:deslime	:cyclone_overflow	:flotation_feed, :deslimed_slime	320
:ROUGHER_FLOTATION	:flotation	:rougher_cells	:flotation	:flotation_feed, :reagent_fatty_acid, :reagent_silicate	:rougher_concentrate, :scavenger_tailings	320
:CLEANER_FLOTATION	:flotation	:cleaner_cells	:flotation	:rougher_concentrate, :reagent_silicate	:cleaner_concentrate, :scavenger_tailings	160
:SCAVENGER_FLOTATION	:flotation	:scavenger_cells	:flotation	:scavenger_tailings, :reagent_fatty_acid	:final_concentrate, :flotation_tailings	192
:CONCENTRATE_THICKENING	:flotation	:concentrate_thickener	:thicken	:cleaner_concentrate, :final_concentrate	:thickener_overflow, :concentrate_cake	144
:CONCENTRATE_FILTRATION	:filtration	:concentrate_filter	:filter_dry	:concentrate_cake, :process_water	:dry_rock, :thickener_overflow	134.4
:FLUORINE_SCRUBBING	:filtration	:fluorine_scrubber	:scrub	:flash_vapour, :wash_water	:fluosilicic_acid, :stack_gas	150
:ATTACK_TANKS	:attack	:attack_tank	:digest	:dry_rock, :strong_sulphuric, :recycle_acid, :dilution_water	:attack_slurry	150
:DIGESTER	:attack	:digester	:digest	:attack_slurry	:digester_slurry	150
:FLASH_COOLING	:attack	:flash_cooler	:flash	:digester_slurry, :cooling_water	:filter_feed, :flash_vapour	150
:FILTRATION	:filtration	:belt_filter	:filter	:filter_feed, :wash_water	:filter_weak_acid, :gypsum_cake	150
:GYPSUM_DEWATERING	:filtration	:gypsum_belt	:dewater	:gypsum_cake, :wash_water	:process_water	150
:SULPHUR_MELTING	:acid_plant	:sulphur_melter	:melt		:molten_sulphur	150
:SULPHUR_BURNING	:acid_plant	:sulphur_burner	:burn	:molten_sulphur, :burner_air, :condensate	:so2_gas, :steam	345.48
:SO2_CONVERSION	:acid_plant	:converter	:convert	:so2_gas, :cooling_water	:so3_gas	345.48
:H2SO4_ABSORPTION	:acid_plant	:absorption_tower	:absorb	:so3_gas, :absorption_water	:strong_sulphuric	345.48
:EVAPORATION	:evaporation	:forced_circulation_evaporator	:evaporate	:filter_weak_acid, :steam	:concentrated_acid, :evaporator_condensate	150
:ACID_STORAGE	:evaporation	:acid_storage_tank	:store	:concentrated_acid	:recycle_acid	150
:SURFACE_CONDENSING	:evaporation	:surface_condenser	:condense	:evaporator_condensate, :cooling_water	:process_water	150
:AMMONIATION	:granulation	:ammonia_sparger	:ammoniate	:concentrated_acid, :ammonia	:dap_melt, :granulator_offgas	150
:GRANULATION	:granulation	:rotary_granulator	:granulate	:dap_melt, :undersize_recycle, :oversize_crush, :process_water	:green_pellets, :granulator_offgas	268
:DRYING	:granulation	:dryer_drum	:dry	:green_pellets, :natural_gas	:dried_pellets, :dryer_offgas	268
:COOLING	:granulation	:cooler_drum	:cool	:dried_pellets, :compressed_air	:cooled_pellets, :cooler_offgas	268
:SIZING	:granulation	:size_screens	:screen	:cooled_pellets	:product_dap, :oversize_crush, :undersize_recycle	268
:OVERSIZE_CRUSHING	:granulation	:crusher_oversize	:crush	:oversize_crush	:undersize_recycle	48.24
:COATING	:granulation	:coating_drum	:coat	:product_dap	:coated_product	268
:PRODUCT_SCRUBBING	:granulation	:product_scrubber	:scrub	:granulator_offgas, :dryer_offgas, :cooler_offgas, :process_water	:scrubber_liquor, :stack_gas	268
:STEAM_TURBINE	:utilities	:steam_turbine	:power	:steam	:electricity, :condensate	4.2
:COOLING_WATER	:utilities	:cooling_tower	:cool	:cooling_water_return	:cooling_water	9,800
:TAILINGS_THICKENING	:tailings	:tailings_thickener	:thicken	:flotation_tailings, :deslimed_slime	:thickener_overflow	192
:WATER_RECOVERY	:tailings	:process_water_pump	:recover	:thickener_overflow	:process_water	150

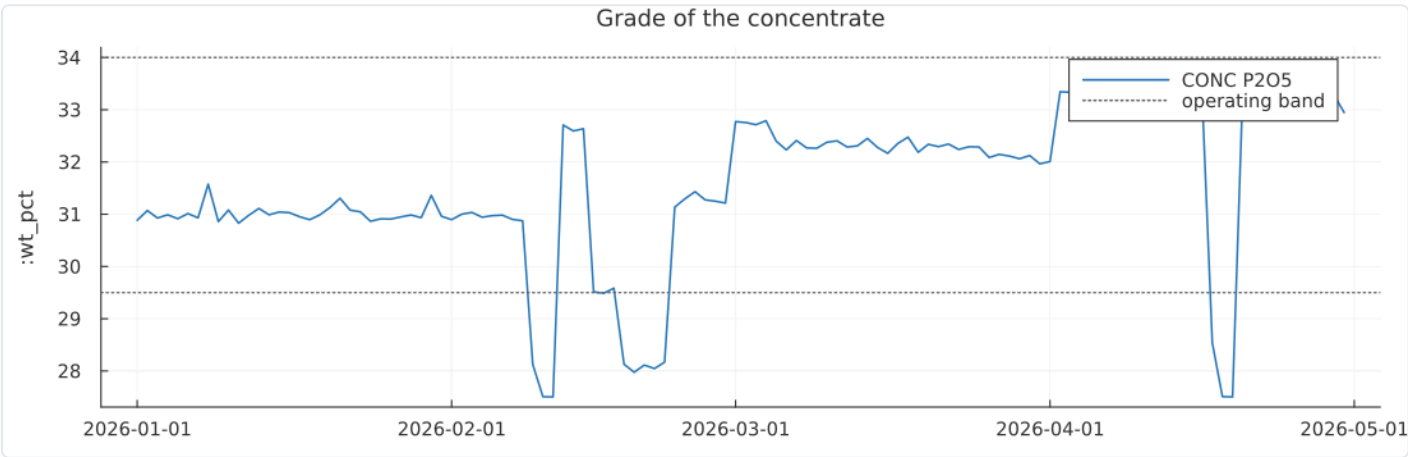
Ore preparation, flotation and the metallurgical balance

ASSAYS OF THE BLENDED ORE BODIES

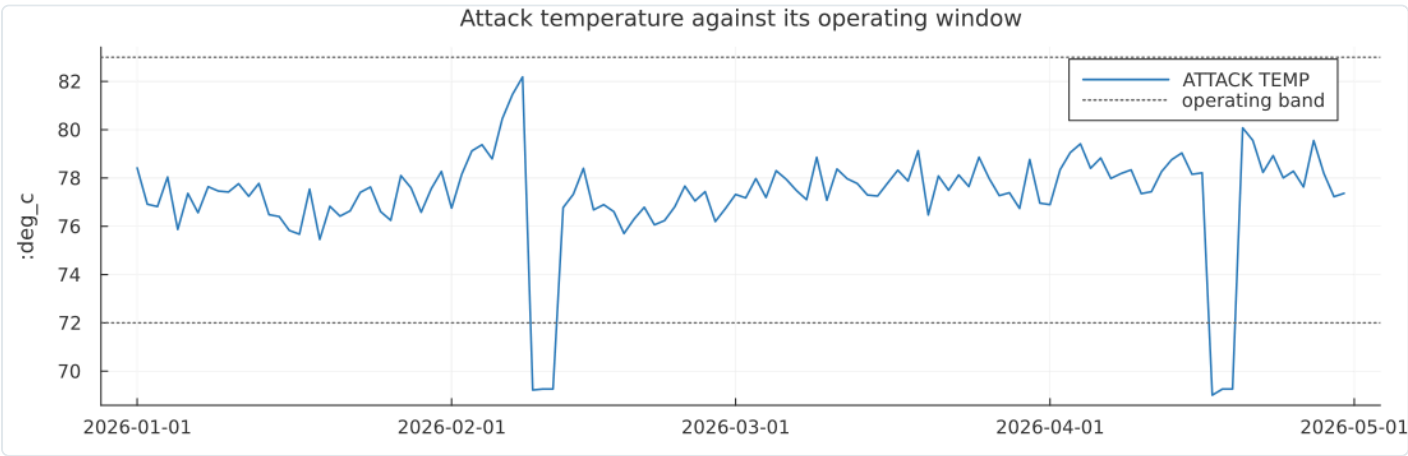
Ore body	P2O5 %	BPL %	CaO %	SiO2 %	Fe2O3 %	Al2O3 %	MgO %	F %	CO2 %	H2O %	CaO:P2O5	Gangue %
:boucraa	34.8	76.04	51	2.1	0.3	0.3	0.4	3.6	4.5	1	1.47	4.6
:florida_pebble	29.8	65.12	45.5	6	1.3	1.2	0.5	3.3	6.5	2	1.53	12.5
:gafsa	29	63.37	47	5.5	0.8	0.9	1.1	3.5	7.5	2.8	1.62	10.8
:jacobina	29.8	65.12	46	5	4.5	2.5	1.5	1.8	5.5	2.5	1.54	16.5
:khouribga	30.9	67.52	50.5	3	0.4	0.4	0.6	3.4	6	1.2	1.63	6.4
:tapira	33.4	72.99	48	4	3	2	1	2	4	1.5	1.44	12

METALLURGICAL BALANCE OF THE CAMPAIGN

Quantity	Value
Ore processed	1,578.6 kt
Flotation feed grade	25.23 % P2O5
Concentrate	1,067.7 kt at 31.78 % P2O5
Flotation recovery	85.18 %
Tailings grade	10.78 % P2O5
Grind P80	153 um
Specific grinding energy	11.38 kWh per t of ore
Reagent	0.94986 kg per t of ore
Two-product balance residual	-0.99366 %



P2O5 of the concentrate against the specification of the attack section

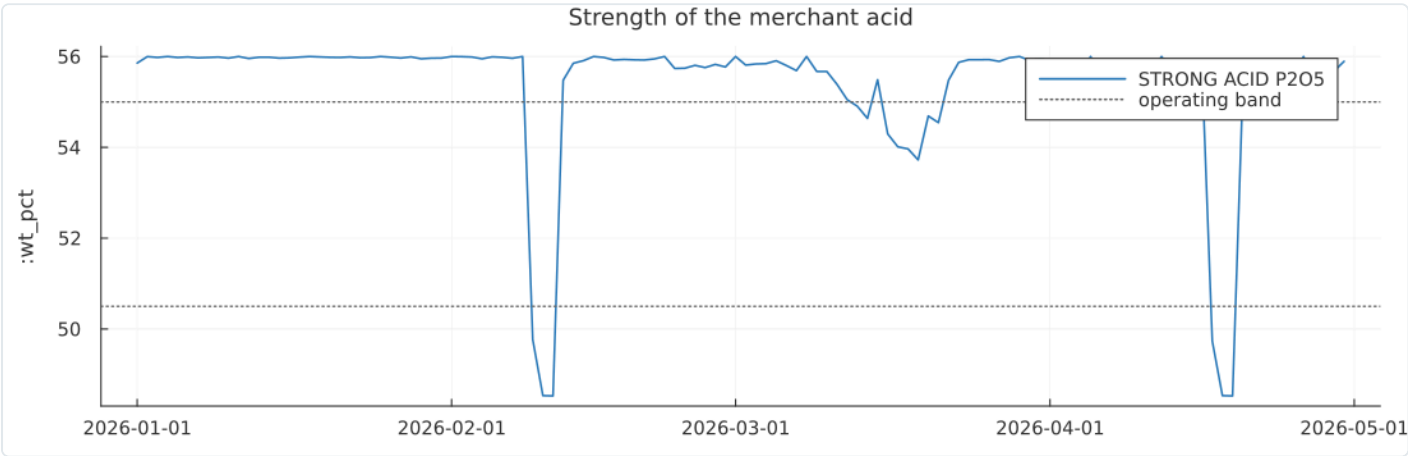


Temperature of the attack tanks, with the crystallisation window of the dihydrate process

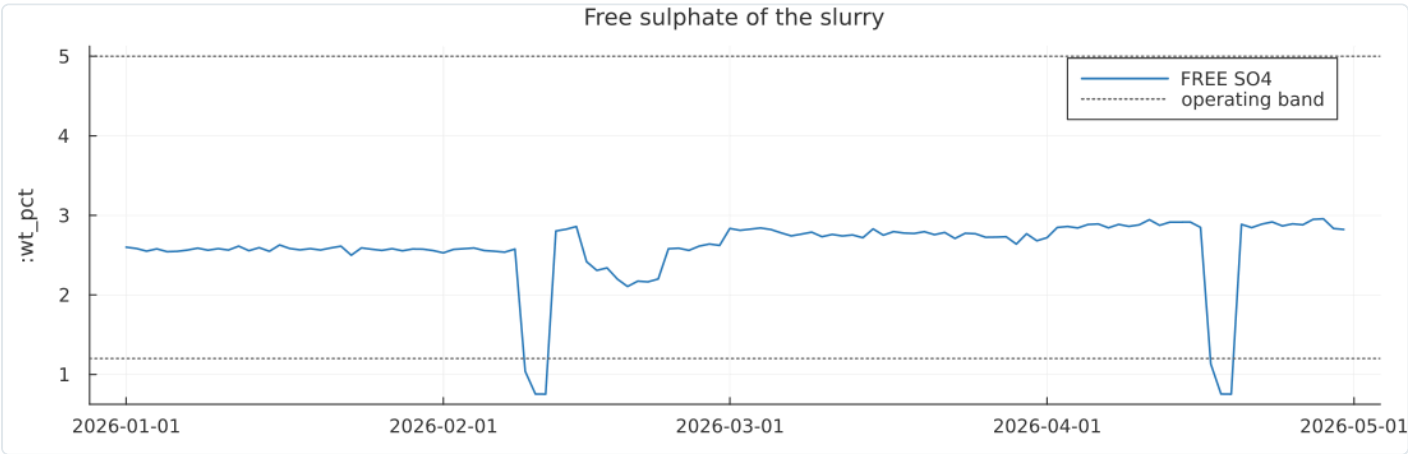
The reaction train: attack, filtration and evaporation

REACTION, FILTRATION AND EVAPORATION

Quantity	Value
P2O5 to the attack tanks	334.5 kt
Dry rock fed	1,051.7 kt
Sulphuric acid	952.3 kt (2.85 t per t P2O5)
Sulphur burnt	310.9 kt (0.9294 t per t P2O5)
Attack temperature	77.7 C
Free sulphate	2.68 wt% of slurry
Gypsum produced	1,566.1 kt (4.68 t per t P2O5)
Filter productivity	8.43 t P2O5 per m2 per day over 320 m2
Water-soluble loss	7.82 kt of P2O5 (2.34 % of the feed) at 0.49912 % free P2O5 in the cake
Filtrate strength	28.77 % P2O5
Evaporator steam	136.3 kt (0.4075 t per t P2O5, economy 4.13)
Merchant acid	142.1 kt at 55.74 % P2O5



P2O5 of the merchant acid against the 52 % specification



Free sulphuric acid of the slurry, the crystallisation window of the gypsum

The acidulation demand of the blend is 2.85 t of H2SO4 per t of P2O5 against a stoichiometric 2.3 plus the registered excess of 2.5 %; the gap is the free carbonate of the ore, which is why the assay of the blend is an operating variable and not a laboratory curiosity.

Phosphoric acid: balance, specification and cost

The acid plant produced 142.1 kt of merchant acid -- 79.2 kt P2O5, 109.4 kt H3PO4 on the 100 % basis -- at 55.74 % P2O5, which is the :technical grade. The evaporators removed 507 thousand t of water with 0.41729 t of steam per t of P2O5 concentrated, an economy of 3.72 t of water per t of steam. The acid leaves the plant **Compliant** against the specification of the traded grade.

THE MERCHANT ACID OF THE CAMPAIGN

Quantity	Value
Merchant acid	142.1 kt (79.2 kt P2O5, 109.4 kt as H3PO4)
Grade	55.74 % P2O5 (76.96 % H3PO4) -- :technical
P2O5 of the attack	the filters receive 24.2 % of it as merchant acid, the rest goes to the granulation line
Yield	the filtration and the attack keep 97.66 % of the P2O5 fed; the evaporation closes at 101.03 %
Concentration	2.83 m3 of feed and 1.16 GJ of steam per t P2O5, 61 kWh of blower and pumps
Cost	506 :USD per t P2O5 as acid (282 per t of acid) against a price of 905 and a margin of 399 per t P2O5

THE MERCHANT ACID AGAINST THE SPECIFICATION OF THE TRADED GRADE

Specification	Instrument	Campaign	Limit	Unit	Direction	Margin %	Status
Colour	:ACID_COLOUR		2.83	4 :count	:le	29.28	Compliant
F	:ACID_F		0.3555	0.5 :wt_pct	:le	28.89	Compliant
Fe al	:ACID_FE_AL		0.195	1 :wt_pct	:le	80.5	Compliant
P2o5	:STRONG_ACID_P2O5		55.74	52 :wt_pct	:ge	7.2	Compliant
So4	:ACID_SO4		1.14	1.5 :wt_pct	:le	23.7	Compliant
Solids	:ACID_SOLIDS		0.2953	0.5 :wt_pct	:le	40.94	Compliant

WHERE THE P2O5 OF THE ATTACK GOES

Destination	t P2O5	Share %	Note
Merchant acid	79,205	23.68	to the acid storage, the grade the customer buys
Fertiliser route	247,454	73.98	to the granulation line as concentrated acid
Gypsum cake	7,816.86	2.34	water-soluble P2O5 that left with the phosphogypsum
Unaccounted	0	0	sampling and rounding of the balance, targets zero

THE STEPS OF THE ACID ROUTE

Step	In (t P2O5)	Out (t P2O5)	Loss (t P2O5)	Yield %	Note
Attack and filtration	334,476	326,659	7,816.86	97.66	the cake carries the free P2O5 out of the circuit
Concentration	326,659	78,398	0	101.03	the closure of the evaporation against the registered merchant split
Merchant acid	78,398	79,205	0	101.03	the 24.2 % of the filtered P2O5 the storage tank receives

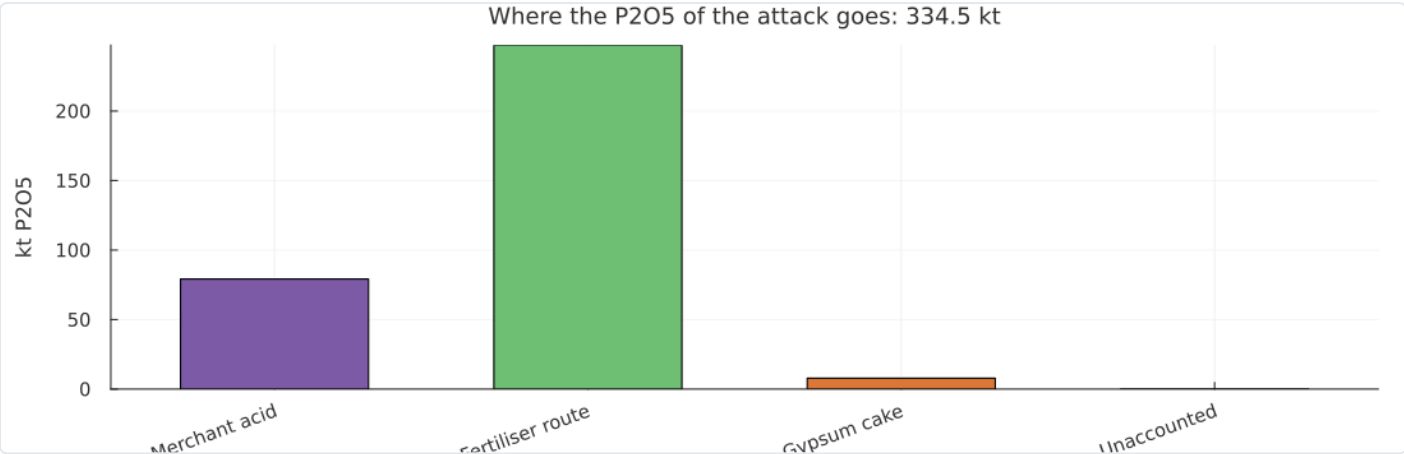
WHAT A TONNE OF P2O5 COSTS IN ROCK, ACID, STEAM AND WATER

Consumption	Value	Unit	Basis	Note
Rock	3.22	:t_ph	:per_p2o5_sold	t of rock per t P2O5 sold as acid, allocated by the acid share
Sulphuric acid	2.92	:t_ph	:per_p2o5_sold	t of H2SO4 per t P2O5 sold as acid
Sulphur	0.9517	:t_ph	:per_p2o5_sold	t of sulphur burnt per t P2O5 sold as acid
Steam	0.4173	:t_ph	:per_p2o5_concentrated	t of steam per t P2O5 the evaporators concentrate, both products included
Electricity	61.2	:kwh_per_t	:per_p2o5_concentrated	kWh per t P2O5 concentrated: blower of the acid plant and evaporator set
Water	2.83	:m3_ph	:per_p2o5_concentrated	m3 of evaporator feed per t P2O5 concentrated
Thermal energy	1.16	:gj_per_t	:per_p2o5_concentrated	GJ of steam per t P2O5 concentrated
Gypsum	4.79	:t_ph	:per_p2o5_sold	t of phosphogypsum per t P2O5

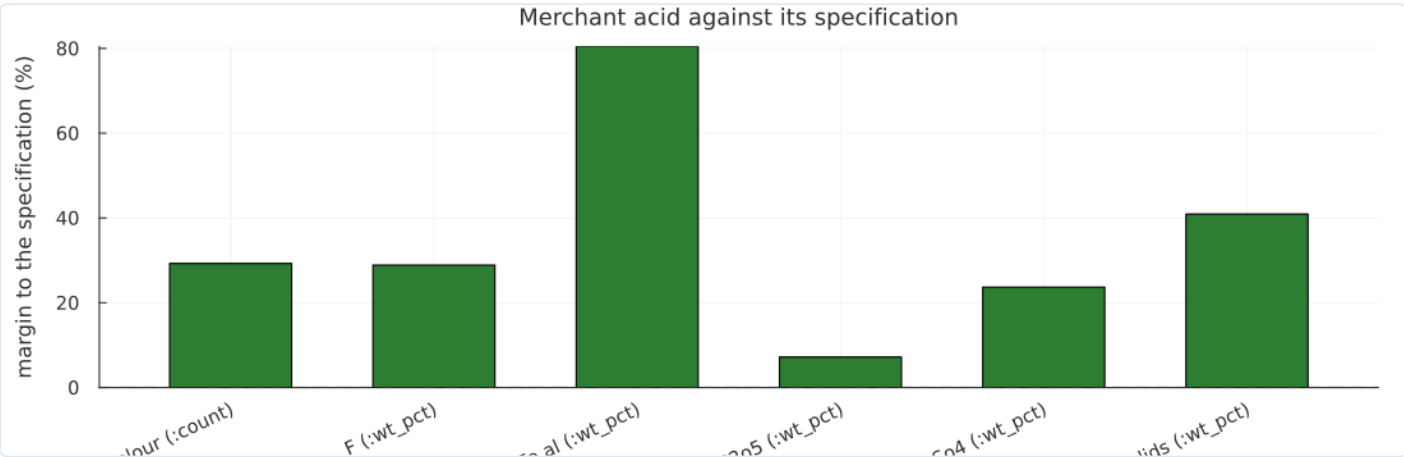
THE TARGETS OF THE ACID ROUTE

Metric	Campaign	Target	Unit	Direction	Margin %	Status
:acid_concentration_yield		101.03	97 :pct	:ge	4.15	Compliant
:acid_solids		0.2953	0.5 :wt_pct	:le	40.94	Compliant
:acid_electricity		61.2	75 :kwh_per_t	:le	18.4	Compliant
:acid_so4		1.14	1.5 :wt_pct	:le	23.7	Compliant
:acid_steam		0.4173	0.62 :t_ph	:le	32.7	Compliant

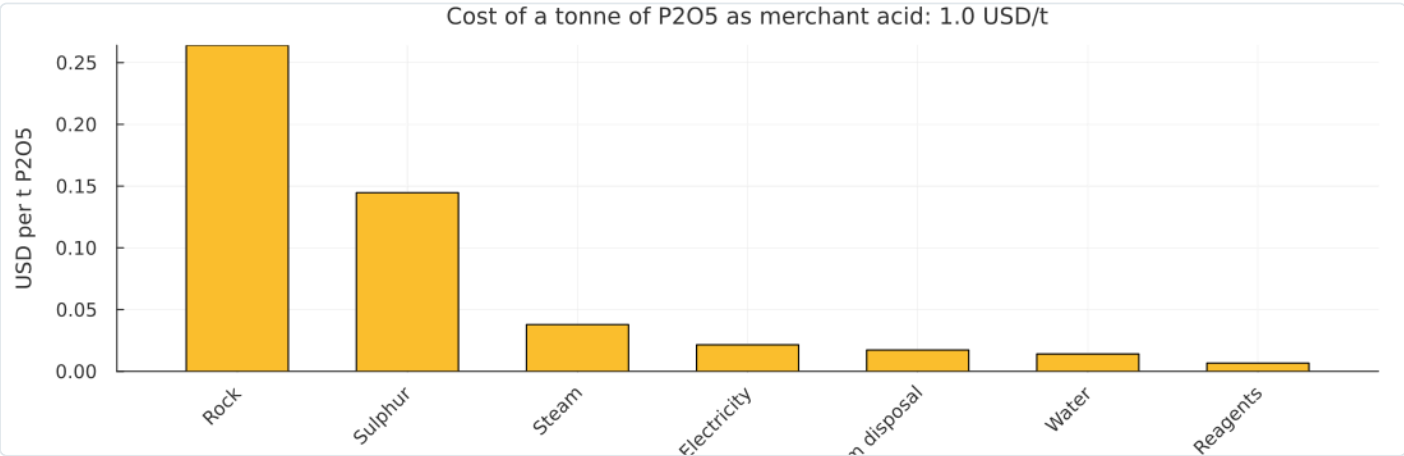
Metric	Campaign	Target	Unit	Direction	Margin %	Status
acid_cost		505.93	600 :usd_per_t	:le	15.68	Compliant
acid_f		0.3555	0.5 :wt_pct	:le	28.89	Compliant
acid_p2o5_yield		97.66	96.5 :pct	:ge	1.21	Compliant
acid_strength		55.74	52 :wt_pct	:ge	7.2	Compliant



Where the P2O5 of the attack goes: the acid route and the fertiliser route



The merchant acid against the specification of the traded grade



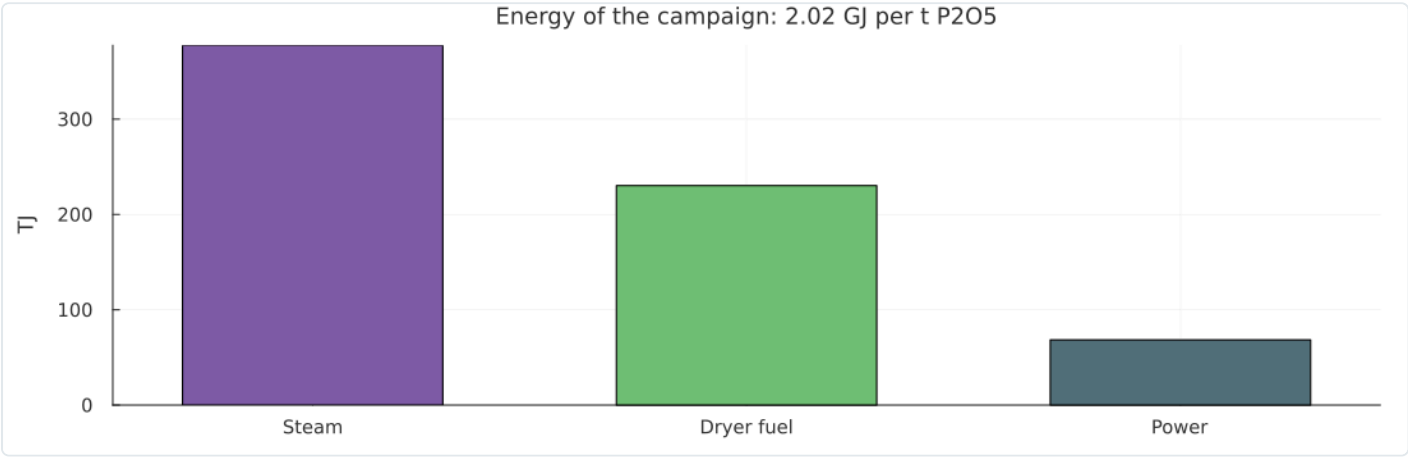
Cost of a tonne of P2O5 as merchant acid, item by item

Acid route: the plant kept 97.66 % of the P2O5 at the filters, sold 142.1 kt of acid at 55.74 % P2O5 for 905 per t P2O5, and spent 506.

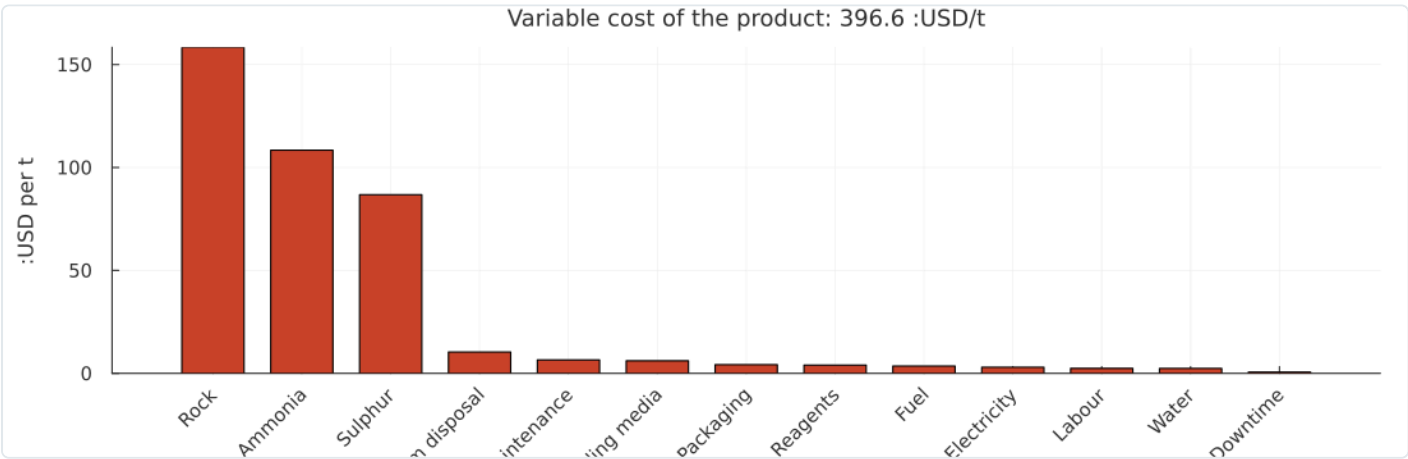
Granulation, product quality, energy, cost and carbon

GRANULATION, PRODUCT AND THE ECONOMIC AND ENVIRONMENTAL VIEW

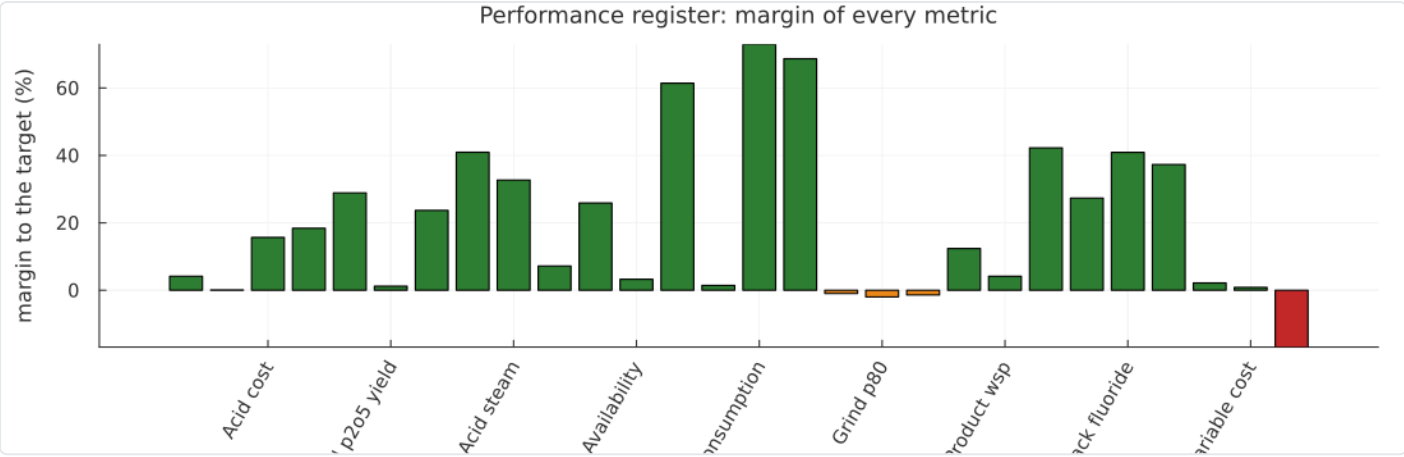
Quantity	Value
Product	544.5 kt of Dap at 46.05 % P2O5
Nitrogen	18.12 % (specification 17.4 to 19.2)
Moisture and solubility	1.75 % H2O and 88.5 % water-soluble P2O5
Ammonia	121.5 kt (0.3631 t per t P2O5, molar ratio 1.51)
Recycle ratio	7 : 1
Override and fines	6.9 % and 28.3 %
Specific energy	2.02 GJ per t P2O5 (56.8 kWh of power)
Dryer fuel	230.3 TJ of :natural_gas
Variable cost	215.95 M:USD (396.6 per t of product)
Revenue and margin	354,821 M:USD of revenue, margin 651,248 per t
Carbon	55.9 kgCO2e per t P2O5 (scope 1 11,658 kt, scope 2 7,046.9 kt)
Stack	2.95 mg F per Nm3 (limit 5) and 21.8 mg dust per Nm3 (limit 30)
Water	1,123.6 thousand m3 (3.36 t per t P2O5)



Where the energy of the campaign goes and the specific energy of the product



The variable cost of the tonne of product, item by item



Margin of every performance metric of the register against its target

The performance register of the campaign

● COMPLIANT

24 metrics

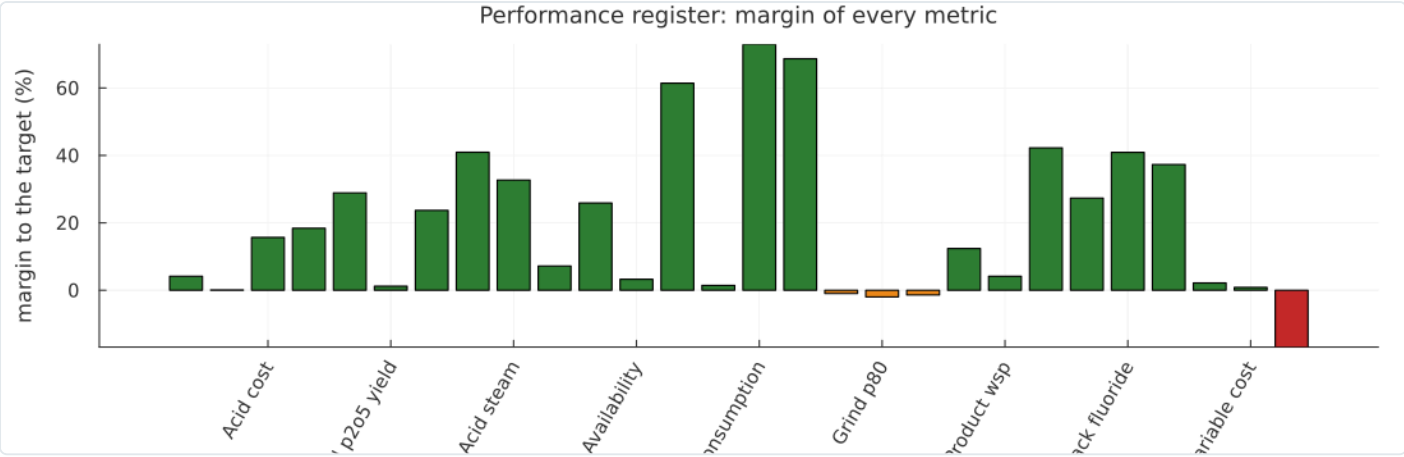
● AT RISK

3 metrics

● NONCOMPLIANT

1 metrics

Metric	Campaign	Target	Unit	Direction	Margin %	Status	Basis
:acid_concentration_yield	101.03	97	:pct	:ge	4.15	Compliant	P2O5 in the merchant acid over the P2O5 the evaporators concentrate: a closure check at three per cent of meter slack
:acid_consumption	2.85	2.85	:t_ph	:le	0.1048	Compliant	t H2SO4 per t P2O5, stoichiometric demand plus the registered excess
:acid_cost	505.93	600	:usd_per_t	:le	15.68	Compliant	variable cost per t P2O5 sold as merchant acid: feed by acid share, concentration in full
:acid_electricity	61.2	75	:kwh_per_t	:le	18.4	Compliant	kWh per t P2O5 concentrated, blower of the acid plant and evaporator set
:acid_f	0.3555	0.5	:wt_pct	:le	28.89	Compliant	fluorine of the merchant acid: the specification limit of the grade
:acid_p2o5_yield	97.66	96.5	:pct	:ge	1.21	Compliant	P2O5 reaching the filters over the P2O5 fed to the attack
:acid_so4	1.14	1.5	:wt_pct	:le	23.7	Compliant	sulphate of the merchant acid: the specification limit of the grade
:acid_solids	0.2953	0.5	:wt_pct	:le	40.94	Compliant	solids of the merchant acid: the specification limit of the grade
:acid_steam	0.4173	0.62	:t_ph	:le	32.7	Compliant	t steam per t P2O5 concentrated, at an economy of 3.0
:acid_strength	55.74	52	:wt_pct	:ge	7.2	Compliant	P2O5 in the merchant acid
:ammonia_consumption	0.3631	0.49	:t_ph	:le	25.89	Compliant	t NH3 per t P2O5 of DAP
:availability	95	92	:pct	:ge	3.26	Compliant	hours the reaction section was available
:carbon_intensity	55.92	145	:kg_ph	:le	61.43	Compliant	kgCO2e per t P2O5, scope 1 and 2
:completeness	99.42	98	:pct	:ge	1.45	Compliant	usable intervals of the historian
:electricity_consumption	56.79	210	:kwh_per_t	:le	72.96	Compliant	kWh per t P2O5, mine to bagged product
:filtration_rate	8.43	5	:t_per_m2_d	:ge	68.68	Compliant	t P2O5 per m2 of filter per day
:flotation_recovery	85.18	86	:pct	:ge	-0.9576	At risk	registered recoveries of the rougher-cleaner circuit
:grind_p80	152.96	150	:um	:le	-1.97	At risk	liberation size of the blend
:plant_p2o5_recovery	82.83	84	:pct	:ge	-1.39	At risk	P2O5 in the product plus the merchant acid over the P2O5 in the flotation feed
:product_moisture	1.75	2	:wt_pct	:le	12.39	Compliant	moisture of the bagged product
:product_wsp	88.53	85	:wt_pct	:ge	4.15	Compliant	water-soluble P2O5 of the product
:specific_energy	2.02	3.5	:gj_per_t	:le	42.25	Compliant	GJ per t P2O5: steam of the evaporators, fuel of the dryers and power
:stack_dust	21.8	30	:count	:le	27.34	Compliant	mg dust per Nm3 at the stack, the permitted limit
:stack_fluoride	2.95	5	:count	:le	40.91	Compliant	mg F per Nm3 at the stack, the permitted limit
:steam_consumption	0.4075	0.65	:t_ph	:le	37.3	Compliant	t steam per t P2O5 fed, weak acid at 28.5 % raised to 52 % at an economy of 3.0
:sulphur_consumption	0.9294	0.95	:t_ph	:le	2.17	Compliant	t sulphur per t P2O5 at 99.7% conversion
:variable_cost	396.6	400	:usd_per_t	:le	0.8495	Compliant	variable cost per tonne of DAP
:water_soluble_loss	2.34	2	:pct	:le	-16.85	Noncompliant	water-soluble P2O5 lost with the gypsum, per cent of the P2O5 fed



Margin of every performance metric of the register against its target

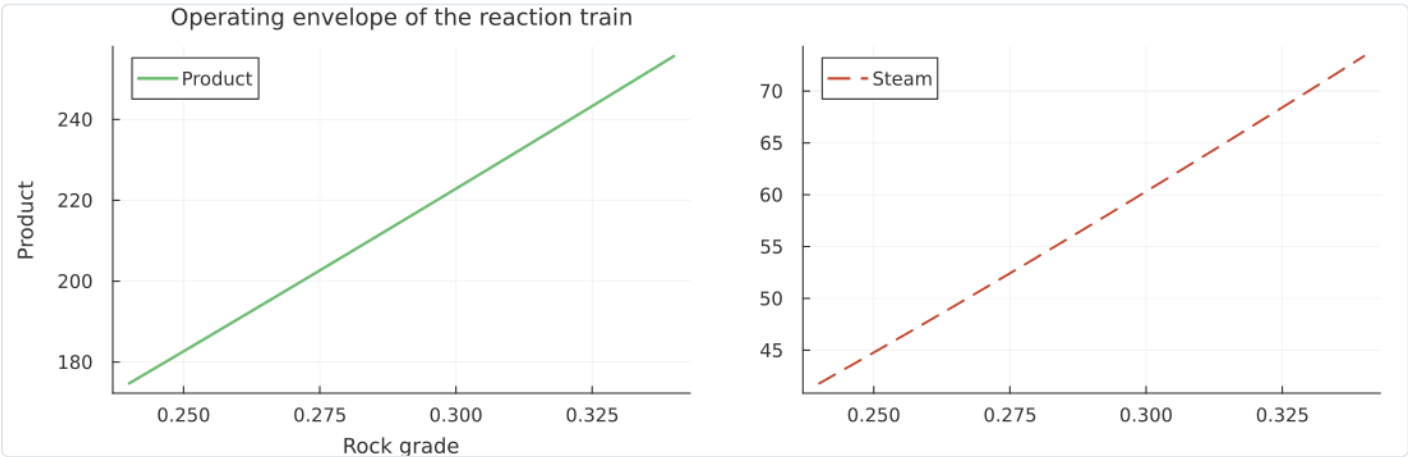
The digital twin: steady state, limits and alignment

The steady-state twin of the reaction train is a square system of 15 nonlinear balances solved with ModelingToolkit (ModelingToolkit nonlinear balance, mtkcompile + NonlinearProblem). It converged in 12.13 s with a largest residual of 0.0 -- the residual is reported because a solution nobody checked is a guess.

Unknown	Value	Unit	Description
:p2o5_fed	150	:t_ph	P2o5 fed
:excess	0.02562	:fraction	Excess
:free_so4	2.6606	:wt_pct	Free so4
:temperature	77.2418	:deg_c	Temperature
:efficiency	0.979187	:fraction	Efficiency
:p2o5_digested	146.878	:t_ph	P2o5 digested
:gypsum	737.2408	:t_ph	Gypsum
:p2o5_acid	142.4545	:t_ph	P2o5 acid
:water_liquor	366.0578	:t_ph	Water liquor
:acid_strength_weak	28.014	:wt_pct	Acid strength weak
:steam	65.1559	:t_ph	Steam
:water_evaporated	234.5613	:t_ph	Water evaporated
:ammonia	51.9605	:t_ph	Ammonia
:product	235.1041	:t_ph	Product
:merchant_p2o5	34.1891	:t_ph	Merchant p2o5

OPERATING WINDOW AT THE SOLUTION

Variable	At the solution	Low	High	Unit	Margin low	Margin high	Status
:acid_strength_weak	28.01	27	33	:wt_pct	1.01	4.99	Compliant
:ammonia	51.96	20	55	:t_ph	31.96	3.04	Compliant
:excess	0.0256	0.015	0.055	:fraction	0.0106	0.0294	Compliant
:free_so4	2.66	1.2	7	:wt_pct	1.46	4.34	Compliant
:steam	65.16	40	130	:t_ph	25.16	64.84	Compliant
:temperature	77.24	72	83	:deg_c	5.24	5.76	Compliant

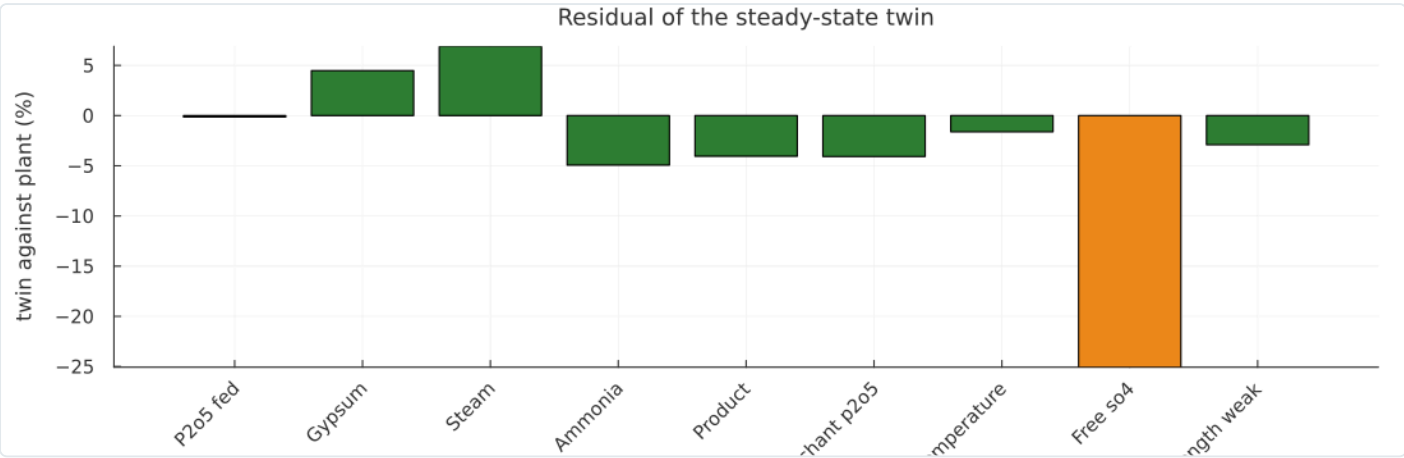


Operating envelope of the reaction train from the steady-state twin

Against the campaign it is supposed to describe, the twin scores 94 % of validity: the residuals below are what a commissioning would absorb as a bias, and the reason they are printed is that a twin is only as good as this table.

Variable	Twin	Plant	Unit	Residual %	Tolerance %	Status
:p2o5_fed	122.08	122.25	:t_ph	-0.1398	10	Compliant
:gypsum	598.05	572.42	:t_ph	4.48	12	Compliant
:steam	53.26	49.82	:t_ph	6.9	25	Compliant
:ammonia	42.2	44.39	:t_ph	-4.94	25	Compliant
:product	190.95	199.01	:t_ph	-4.05	25	Compliant
:merchant_p2o5	27.77	28.95	:t_ph	-4.08	20	Compliant

Variable	Twin	Plant	Unit	Residual %	Tolerance %	Status
:temperature	76.43	77.69	:deg_c	-1.62	5	Compliant
:free_so4	2.01	2.68	:wt_pct	-25.07	15	At risk
:acid_strength_weak	27.93	28.77	:wt_pct	-2.91	8	Compliant



Residual of the steady-state twin against the campaign it describes

The dynamic twin and the state estimation

The dynamic twin integrates 11 states with `OrdinaryDiffEqTsit5.Tsit5{typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!), FastBroadcast.Serial}(OrdinaryDiffEqCore.trivial_limiter!, OrdinaryDiffEqCore.trivial_limiter!, FastBroadcast.Serial())` on the hourly clock of the plant, from the same correlations the historian uses: the holdup of the attack tanks, the strength of the acid, the moisture of the granulator bed, the holdup of the mills and the fouling of the evaporator tubes.

DYNAMIC TRAJECTORY OF THE TWIN

Quantity	Value
Horizon	0.0 to 48.0 h
Status	compliant (Success)
P2O5 fed over the horizon	3,934.9 t
Liquor of the attack tanks	306 t to 306 t (-0.0009 %)
Highest temperature	78.02 C
Weakest acid	51.6 % P2O5
Fouling at the end	0.629
Integration time	4.62 s

The Kalman filter of the twin walked the historian for 336 intervals and estimated the states nobody measures. The innovation of every measurement is the table below: it is the residual the controller sees, and a sensor that drifts shows up here first.

Measurement	RMSE of the innovation	Bias	Intervals
:acid_strength	2.21	-2.15	336
:acid_strength_weak	40.48	40.1	336
:bed_moisture	1.37	1.34	336
:filter_rate	58.93	-58.8	336
:fouling_pct	0	0	336
:free_so4	27.28	-26.88	336
:granulator_temperature	1.04	-0.6037	336
:mill_inventory	142.33	-140.42	336
:mill_power	1,874.03	-1,847.63	336
:p80	51.83	-50.83	336
:product_rate	1,283.84	-1,281.18	336
:temperature	15.96	-0.7315	336

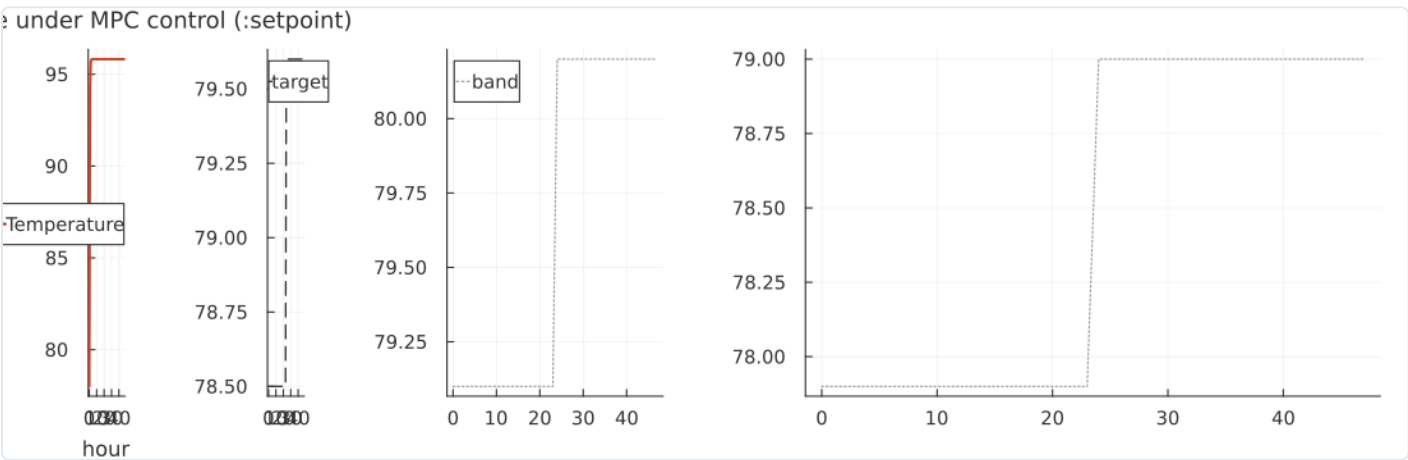
Predictive control of the reaction train

The controller is the one the plant would run: a linear predictive controller over the discrete model 11 x 11 obtained by linearising the ODE twin at its operating point (12 prediction steps, 4 control steps, sample time 1 h), with a Kalman filter estimating the states from the measurements. The plant of the test is the *nonlinear* twin, so the controller never sees the equations it is controlling.

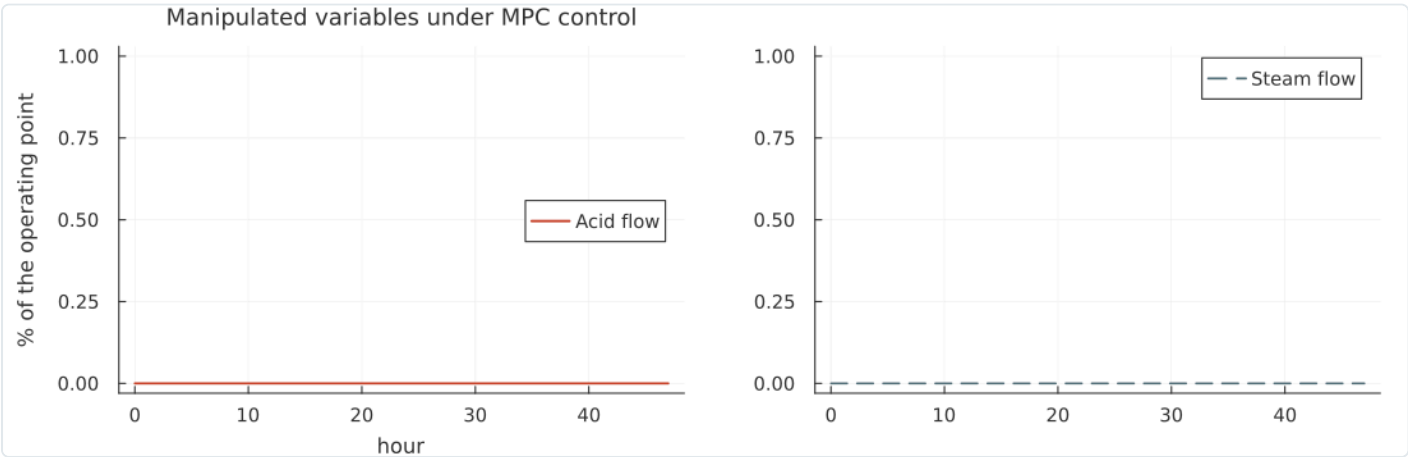
CLOSED LOOP OF THE TWO CONTROLLERS

Quantity	Value
Scenario	:setpoint over 48 hours
Model	11 states, 6 inputs, 12 outputs, Ts = 1 h
Controllability	conditioning 3.4124288612524188e16
Poles	largest magnitude 1
MPC error	14,863 bands and 284 violations
PI error	2,867.68 bands and 233 violations

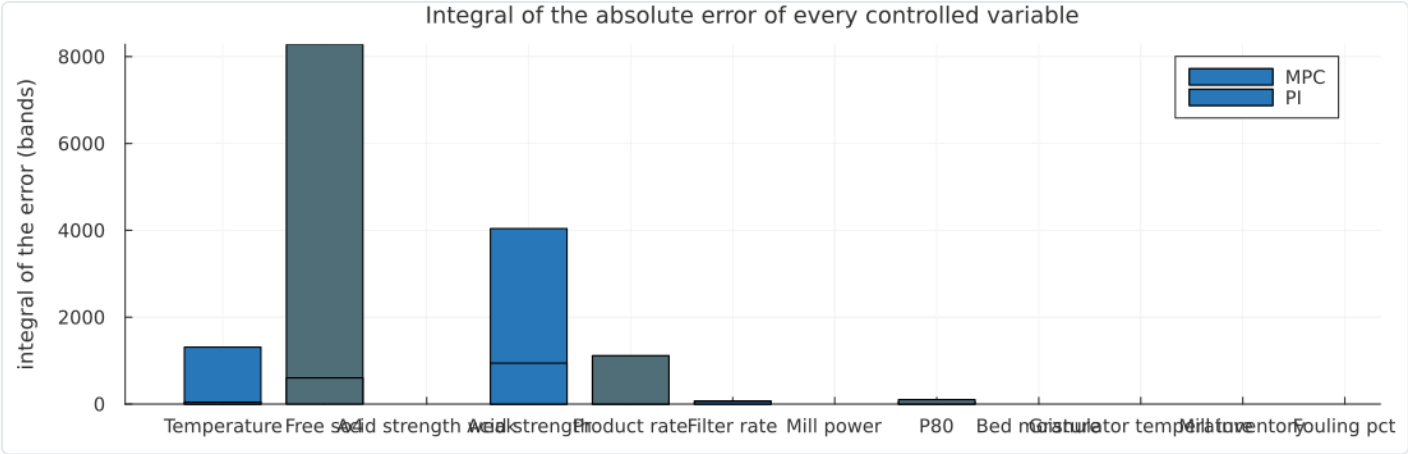
Controlled variable	Band	MPC error	PI error	Improvement %	MPC worst	PI worst	MPC hours out	PI hours out
:temperature	0.6	1,310.92	41.86	-3,031.32	28.87	0.9654	47	0
:free_so4	0.2	8,282.93	604.12	-1,271.06	195.69	17.27	47	42
:acid_strength_weak	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
:acid_strength	0.15	4,041	942.67	-328.68	89.91	23.24	48	48
:product_rate	6	1,102.69	1,115.28	1.13	23.35	23.35	48	48
:filter_rate	0.35	70.65	60.84	-16.12	1.48	1.27	48	48
:mill_power	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
:p80	6	54.82	102.9	46.73	1.17	2.19	46	47
:bed_moisture	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
:granulator_temperature	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
:mill_inventory	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
:fouling_pct	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a



Closed loop of the predictive controller



Manipulated variables the predictive controller used



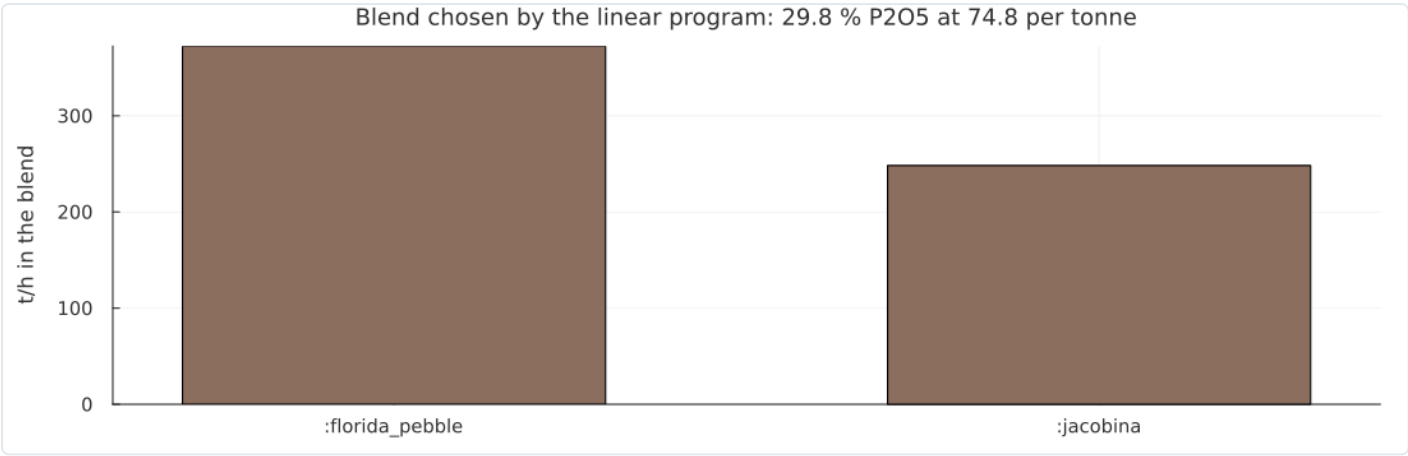
Integral of the absolute error of the two controllers, output by output

WORK THE TWO CONTROLLERS ASKED OF THE PLANT

Manipulated variable	MPC movement %	PI movement %	MPC reversals	PI reversals	MPC range %	PI range %	
:acid_flow		0	14.94	0	0	0	14.94
:ammonia_flow		0	0	0	0	0	0
:cooling_water		0	40	0	0	0	40
:mill_feed		0	2.62	0	0	0	2.62
:steam_flow		0	17.75	0	0	0	17.75
:wash_water		0	0	0	0	0	0

Optimisation of the blend, the operating point and the sourcing

Three models of the same plant are solved every day: the blend of the ore the digester is fed -- a linear program over the registered assays with the specifications of the reaction section --, the operating point of the reaction train -- a nonlinear program over the correlations of the twin -- and the annual sourcing plan, a mixed-integer program because a body is bought by the shipload.



Ore blend the optimiser chose

OPTIMAL BLEND (HIGHS, STATUS :COMPLIANT, 29.8 % P2O5 AT 74.8 PER T)

Ore body	t/h	Share %	Price	P2O5 %	CaO:P2O5	Fe+Al %	MgO %
:florida_pebble	372.41	60	78	29.8	1.53	2.5	0.5
:jacobina	248.28	40	70	29.8	1.54	7	1.5
:boucraa	0	0	96	34.8	1.47	0.6	0.4
:gafsa	0	0	74	29	1.62	1.7	1.1
:khouribga	0	0	82	30.9	1.63	0.8	0.6
:tapira	0	0	88	33.4	1.44	5	1

WHAT ONE MORE UNIT OF EACH SPECIFICATION IS WORTH

Specification	Shadow price
:mgo	-800
:fe_al	-0
:p2o5	0
:sio2	-0
:cao_p2o5	-0
:co2	-0

OPERATING POINT (IPOPT, STATUS :COMPLIANT, MARGIN 182.8 PER T)

Nature	Variable	Optimum
:decision	:recycle_acid_flow	450
:decision	:cooling_water_flow	248
:decision	:steam_flow	80
:decision	:rock_feed	523.81
:decision	:wash_water_flow	220.19
:decision	:ammonia_flow	56.48
:decision	:sulphuric_acid_flow	481.6
:controlled	:product_moisture	1.46
:controlled	:acid_strength_weak	27
:controlled	:gypsum_t	668.07
:controlled	:sulfate_ratio	2.3
:controlled	:p2o5_product	122.92
:controlled	:steam_required	80
:controlled	:product_wsp	88.29
:controlled	:gypsum_free_p2o5	0.5241
:controlled	:acid_excess	1.5
:controlled	:reactor_temperature	80.17
:controlled	:p2o5_fed	166.36

Nature	Variable	Optimum
:controlled	:filtration_rate	12.39
:controlled	:product_n	17.4
:controlled	:acid_strength	53.44

ANNUAL SOURCING PLAN (HIGHS, 72 BINARY DECISIONS, 253.45 PER T P2O5)

Ore body	t per year	Share %	Price	Cost	Months	P2O5 %
:gafsa	1.283079e6	27.8	74	9.4947875e7	4	29
:florida_pebble	1.248634e6	27.05	78	9.7393476e7	4	29.8
:jacobina	1.248634e6	27.05	70	8.7404401e7	4	29.8
:tapira	835,538	18.1	88	7.3527364e7	3	33.4
:boucraa	0	0	96	0	0	34.8
:khouribga	0	0	82	0	0	30.9

The inferential sensors: process, vision and acoustics

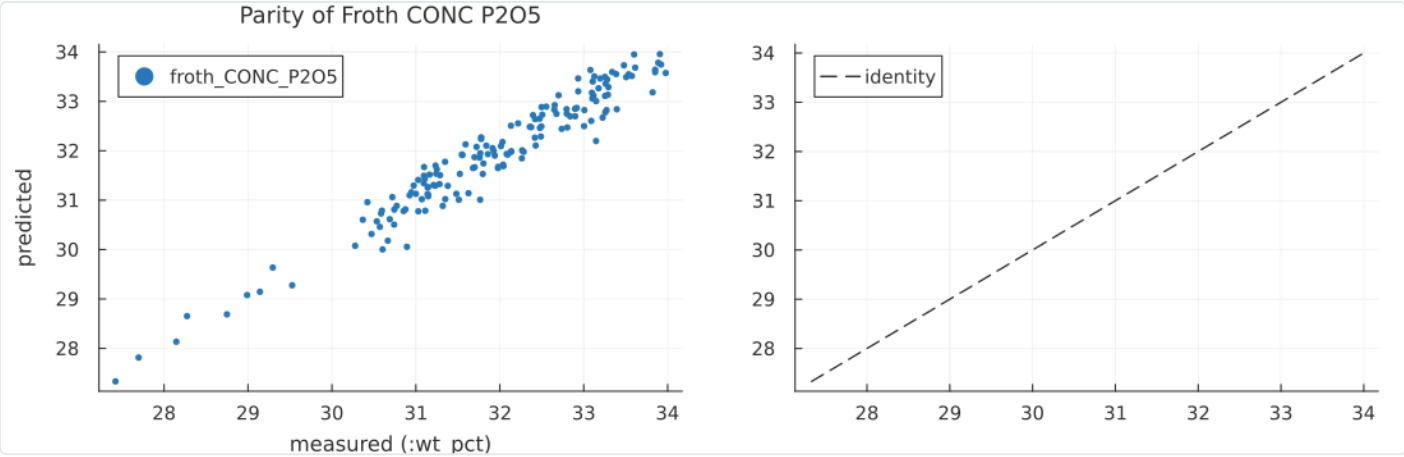
THE REGISTER OF THE INFERENTIAL SENSORS

Quantity	Value									
Sensors	14 inferential models, of which 4 read an image or a sound									
Median R2	0.54428									
Best sensor	:pump_PUMP_VIBRATION									
Status	:at_risk									

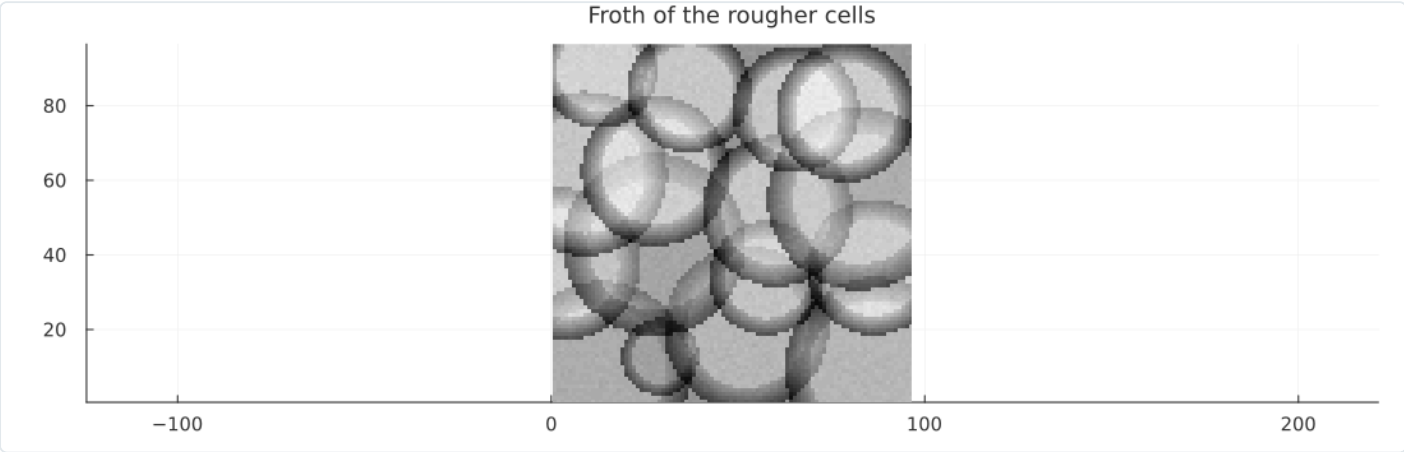
Sensor	Infers	Modality	Unit	R2	RMSE	Bias	In domain %	Drift	Status	Inputs
:pump_PUMP_VIBRATION	:PUMP_VIBRATION	:acoustic	:count	0.99279	0.11026	0.0078	n/a	0	Healthy	rms, crest, kurtosis, band_low, band_mid, band_high, band_veryhigh, centroid, dominant_hz, cavitation_index, tonal_ratio
:froth_CONC_P2O5	:CONC_P2O5	:visual	:wt_pct	0.94903	0.284063	-0.0127	n/a	0	Healthy	img_mean, img_std, img_entropy, img_band_low, img_band_mid, img_band_high, img_edge_density, img_local_contrast, img_bubble_count, img_bubble_px, img_dominant_f
:mill_MILL_POWER	:MILL_POWER	:acoustic	:kw	0.9311	89.0989	-33.98	n/a	0	Healthy	rms, crest, kurtosis, band_low, band_mid, band_high, band_veryhigh, centroid, dominant_hz, cavitation_index, tonal_ratio
:process_STEAM_FLOW	:STEAM_FLOW	:process	:t_ph	0.91184	1.9597	0.0542	100	0.0217	Healthy	EVAP_FEED_FLOW, STRONG_ACID_P2O5, EVAP_VACUUM
:process_CONC_P2O5	:CONC_P2O5	:process	:wt_pct	0.89048	0.456288	0.0048	100	0.0019	Healthy	FLOT_FEED_P2O5, CYCLONE_P80, REAGENT_FLOW
:process_MILL_POWER	:MILL_POWER	:process	:kw	0.80293	202.537	-11.22	100	-4.49	Healthy	ROM_FEED, SLURRY_DENSITY, MILL_FEED_WATER
:process_WEAK_ACID_P2O5	:WEAK_ACID_P2O5	:process	:wt_pct	0.58554	0.203578	-0.02	100	-0.008	Healthy	SLURRY_SG, ATTACK_TEMP, FREE_SO4, WASH_WATER_FLOW
:process_STRONG_ACID_P2O5	:STRONG_ACID_P2O5	:process	:wt_pct	0.50303	0.316116	0.01	100	0.004	Healthy	EVAP_TEMP, STEAM_FLOW, EVAP_VACUUM
:process_CYCLONE_P80	:CYCLONE_P80	:process	:um	0.44705	9.5032	0.592	100	0.2368	Healthy	MILL_POWER, SLURRY_DENSITY, CYCLONE_PRESSURE, MILL_FEED_WATER
:process_PRODUCT_N	:PRODUCT_N	:process	:wt_pct	0.27181	0.292159	-0.0164	100	-0.0066	Healthy	AMMONIA_FLOW, PRODUCT_FLOW, MELT_DENSITY
:process_FILTER_RATE	:FILTER_RATE	:process	:t_per_m2_d	0.23443	1.1356	0.0196	100	0.0079	Healthy	FILTER_VACUUM, WEAK_ACID_P2O5, SLURRY_SG, GYPSUM_FREE_P2O5
:rock_CYCLONE_P80	:CYCLONE_P80	:visual	:um	0.17152	9.0556	0.4531	n/a	0	Degraded	img_mean, img_std, img_entropy, img_band_low, img_band_mid, img_band_high, img_edge_density, img_local_contrast, img_bubble_count, img_bubble_px, img_dominant_f
:process_GYPSUM_FREE_P2O5	:GYPSUM_FREE_P2O5	:process	:wt_pct	0.05478	0.024921	-0.0003	100	-0.0001	Healthy	FILTER_RATE, FILTER_VACUUM, WASH_WATER_FLOW, WEAK_ACID_P2O5
:process_ATTACK_TEMP	:ATTACK_TEMP	:process	:deg_c	-0.00493	16.2509	-0.0224	100	-0.009	Healthy	H2SO4_FLOW, ROCK_FEED, RECYCLE_ACID_FLOW, SLURRY_SG

WHAT THE SENSORS SAY AGAINST WHAT THE PLANT MEASURED

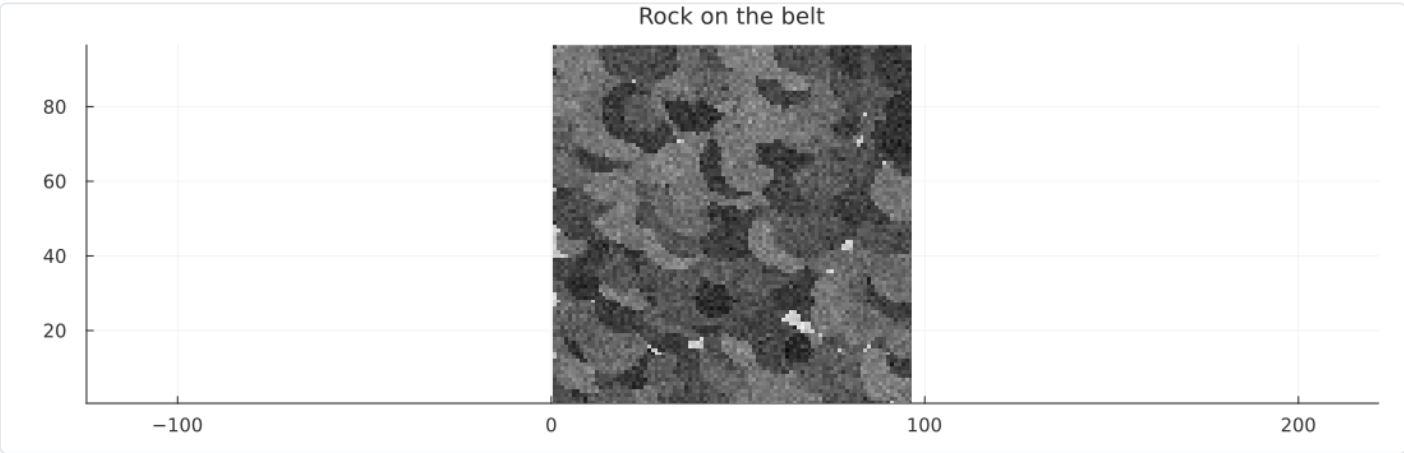
Sensor	Infers	Unit	Estimate	Measured	Difference %	RMSE	Status
:process_STRONG_ACID_P2O5	:STRONG_ACID_P2O5	:wt_pct	55.74	55.89	-0.269	0.3161	Compliant
:process_WEAK_ACID_P2O5	:WEAK_ACID_P2O5	:wt_pct	28.77	28.72	0.1929	0.2036	Compliant
:process_GYPSUM_FREE_P2O5	:GYPSUM_FREE_P2O5	:wt_pct	0.5012	0.5101	-1.76	0.0249	Compliant
:process_CYCLONE_P80	:CYCLONE_P80	:um	151.21	148.3	1.96	9.5	Compliant
:process_MILL_POWER	:MILL_POWER	:kw	5,453.54	5,489.39	-0.653	202.54	Compliant
:process_PRODUCT_N	:PRODUCT_N	:wt_pct	18.14	18.2	-0.327	0.2922	Compliant
:process_FILTER_RATE	:FILTER_RATE	:t_per_m2_d	8.54	8.92	-4.18	1.14	Compliant
:process_CONC_P2O5	:CONC_P2O5	:wt_pct	32.85	32.95	-0.2797	0.4563	Compliant
:process_ATTACK_TEMP	:ATTACK_TEMP	:deg_c	77.88	77.37	0.6672	16.25	Compliant
:process_STEAM_FLOW	:STEAM_FLOW	:t_ph	52.22	52.41	-0.3654	1.96	Compliant



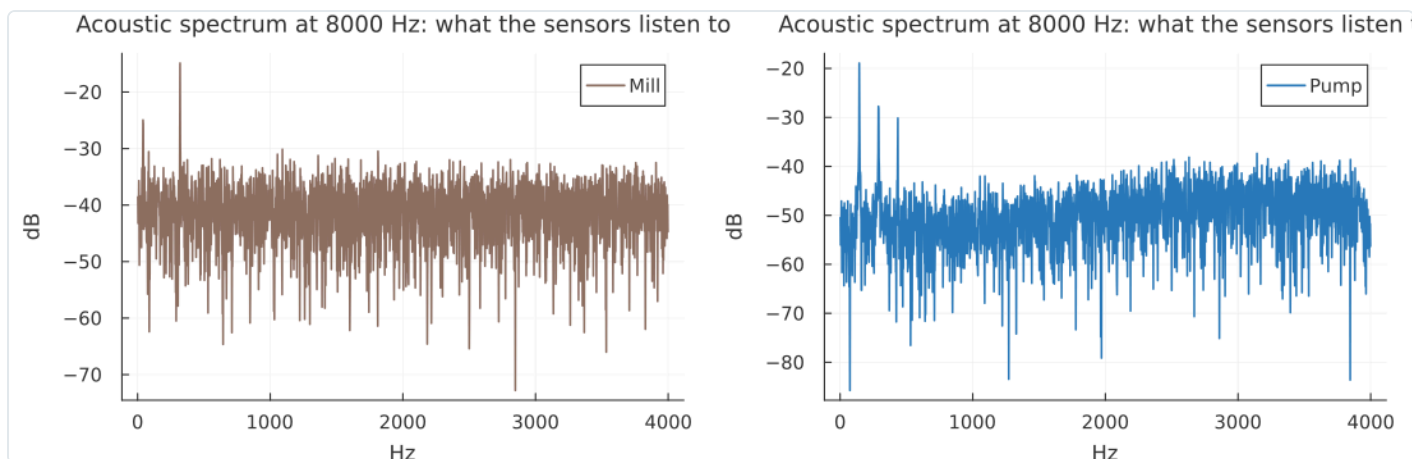
Parity of the froth sensor against the analyser



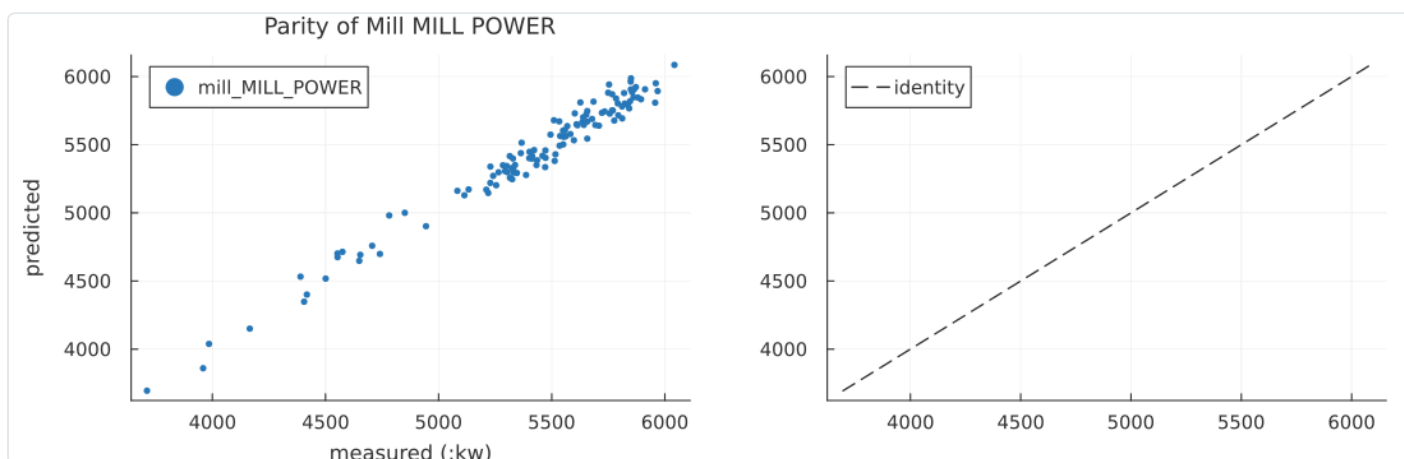
The froth image the visual sensor reads the grade from



The image of the rock the visual sensor reads the grind from



Spectra of the mill and of the pump: the two acoustic channels of the plant



Parity of the acoustic sensor of the mill against the power it draws

Fault detection and diagnostics

DIAGNOSTIC SUMMARY

Quantity	Value
Findings	63 (31 critical, 27 warning, 5 info)
Value at stake	7.7162261e7 :USD a year
Hours in alarm	7,465 h
Rules applied	20 rules of the catalogue
Deviations caught	9 of 15 (60 %)
Status	:action_required

Severity	Area	Rule	Signal	Measured	Reference	Unit	Deviation %	From	To	Hours	Value at stake
Critical	:flotation	:concentrate_grade_low	:CONC_P2O5	27.71	29.5	:wt_pct	6.08	2026-02-09T09:00:00	2026-02-12T03:00:00	67	5.430303e6
Critical	:flotation	:concentrate_grade_low	:CONC_P2O5	27.73	29.5	:wt_pct	6.01	2026-04-17T11:00:00	2026-04-20T03:00:00	65	5.365979e6
Critical	:flotation	:concentrate_grade_low	:CONC_P2O5	28.18	29.5	:wt_pct	4.47	2026-02-18T00:00:00	2026-02-23T04:00:00	125	3.994502e6
Critical	:evaporation	:acid_strength_low	:STRONG_ACID_P2O5	48.73	50.5	:wt_pct	3.51	2026-04-17T12:00:00	2026-04-20T03:00:00	64	1.157993e6
Critical	:evaporation	:acid_strength_low	:STRONG_ACID_P2O5	48.73	50.5	:wt_pct	3.5	2026-02-09T12:00:00	2026-02-12T03:00:00	64	1.153677e6
Critical	:attack	:sulfate_deficit	:FREE_SO4	0.786	1.2	:wt_pct	34.5	2026-02-09T10:00:00	2026-02-12T00:00:00	63	1.00262e6
Critical	:attack	:sulfate_deficit	:FREE_SO4	0.7887	1.2	:wt_pct	34.28	2026-04-17T11:00:00	2026-04-20T00:00:00	62	996,058
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.29	5	:count	5.72	2026-03-31T09:00:00	2026-04-01T00:00:00	16	549,549
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.29	5	:count	5.72	2026-03-29T13:00:00	2026-03-29T21:00:00	9	549,476
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.27	5	:count	5.41	2026-03-28T09:00:00	2026-03-28T20:00:00	12	518,896
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.25	5	:count	4.99	2026-03-22T09:00:00	2026-03-23T15:00:00	31	479,169
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.25	5	:count	4.99	2026-03-18T09:00:00	2026-03-18T15:00:00	7	478,949
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.23	5	:count	4.67	2026-03-19T13:00:00	2026-03-21T05:00:00	41	447,989
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.23	5	:count	4.53	2026-03-24T03:00:00	2026-03-25T05:00:00	27	434,853
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.22	5	:count	4.47	2026-04-01T12:00:00	2026-04-01T23:00:00	12	429,300
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.22	5	:count	4.47	2026-03-27T13:00:00	2026-03-28T07:00:00	19	429,169
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.21	5	:count	4.14	2026-03-21T13:00:00	2026-03-22T01:00:00	13	397,479
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.2	5	:count	3.94	2026-03-17T05:00:00	2026-03-18T05:00:00	25	378,601
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.19	5	:count	3.86	2026-03-27T03:00:00	2026-03-27T09:00:00	7	370,939
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.17	5	:count	3.42	2026-03-30T20:00:00	2026-03-31T07:00:00	12	328,297
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.17	5	:count	3.31	2026-03-18T17:00:00	2026-03-19T08:00:00	16	318,189
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.13	5	:count	2.5	2026-03-25T07:00:00	2026-03-25T11:00:00	5	240,086
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.12	5	:count	2.39	2026-03-25T18:00:00	2026-03-26T00:00:00	7	229,691
Critical	:flotation	:concentrate_grade_low	:CONC_P2O5	29.44	29.5	:wt_pct	0.1968	2026-02-16T05:00:00	2026-02-16T14:00:00	10	175,718
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.08	5	:count	1.65	2026-03-26T02:00:00	2026-03-26T08:00:00	7	158,104
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.08	5	:count	1.52	2026-03-26T10:00:00	2026-03-26T20:00:00	11	145,473
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.07	5	:count	1.49	2026-03-29T02:00:00	2026-03-29T08:00:00	7	142,677
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.07	5	:count	1.44	2026-03-30T02:00:00	2026-03-30T04:00:00	3	138,237
Critical	:granulation	:product_nitrogen_low	:PRODUCT_N	16.4	17.4	:wt_pct	5.78	2026-04-17T08:00:00	2026-04-20T06:00:00	71	43,773
Critical	:granulation	:product_nitrogen_low	:PRODUCT_N	16.4	17.4	:wt_pct	5.76	2026-02-09T07:00:00	2026-02-12T06:00:00	72	43,687
Critical	:granulation	:product_nitrogen_low	:PRODUCT_N	16.98	17.4	:wt_pct	2.4	2026-04-01T08:00:00	2026-04-06T04:00:00	117	18,178
Warning	:filtration	:filter_rate_low	:FILTER_RATE	3.3	4	:t_per_m2_d	17.43	2026-04-18T13:00:00	2026-04-20T00:00:00	36	1.367956e7
Warning	:filtration	:filter_rate_low	:FILTER_RATE	3.34	4	:t_per_m2_d	16.41	2026-02-09T12:00:00	2026-02-12T00:00:00	61	1.2875267e7
Warning	:filtration	:filter_rate_low	:FILTER_RATE	3.53	4	:t_per_m2_d	11.76	2026-04-17T16:00:00	2026-04-18T04:00:00	13	9.232416e6
Warning	:beneficiation	:grind_coarse	:CYCLONE_P80	192.02	185	:um	3.79	2026-02-15T19:00:00	2026-02-23T03:00:00	177	3.185196e6

Severity	Area	Rule	Signal	Measured	Reference	Unit	Deviation %	From	To	Hours	Value at stake
Warning	:granulation	:scrubber_ph_low	:SCRUBBER_PH	4.81	5.6	:count	14.13	2026-04-17T06:00:00	2026-04-20T02:00:00	69	1.139689e6
Warning	:granulation	:scrubber_ph_low	:SCRUBBER_PH	4.81	5.6	:count	14.08	2026-02-09T04:00:00	2026-02-12T02:00:00	71	1.135103e6
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,502.55	4,600	:kw	19.62	2026-03-16T01:00:00	2026-04-05T22:00:00	502	613,734
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,425.29	4,600	:kw	17.94	2026-04-27T01:00:00	2026-04-30T23:00:00	95	561,200
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,422.65	4,600	:kw	17.88	2026-04-20T13:00:00	2026-04-26T20:00:00	152	559,404
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,417.7	4,600	:kw	17.78	2026-02-23T10:00:00	2026-03-01T21:00:00	156	556,038
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,415.38	4,600	:kw	17.73	2026-01-12T01:00:00	2026-01-18T21:00:00	165	554,460
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,414.21	4,600	:kw	17.7	2026-01-01T05:00:00	2026-01-04T21:00:00	89	553,661
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,409.23	4,600	:kw	17.59	2026-02-02T01:00:00	2026-02-08T21:00:00	165	550,279
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,405.43	4,600	:kw	17.51	2026-03-09T02:00:00	2026-03-15T21:00:00	164	547,693
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,403.94	4,600	:kw	17.48	2026-01-05T02:00:00	2026-01-11T21:00:00	164	546,677
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,402.53	4,600	:kw	17.45	2026-04-06T00:00:00	2026-04-12T20:00:00	165	545,721
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,400.59	4,600	:kw	17.4	2026-01-26T02:00:00	2026-02-01T21:00:00	164	544,404
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,398.51	4,600	:kw	17.36	2026-03-02T01:00:00	2026-03-08T20:00:00	164	542,985
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,389.12	4,600	:kw	17.15	2026-01-19T01:00:00	2026-01-25T21:00:00	165	536,601
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,381.26	4,600	:kw	16.98	2026-04-13T02:00:00	2026-04-17T11:00:00	106	531,255
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,366.88	4,600	:kw	16.67	2026-02-12T13:00:00	2026-02-15T12:00:00	72	521,477
Warning	:beneficiation	:slurry_density_high	:SLURRY_DENSITY	44.42	44	:wt_pct	0.9572	2026-03-21T11:00:00	2026-03-30T23:00:00	229	191,234
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	4,787.19	4,600	:kw	4.07	2026-02-16T17:00:00	2026-02-22T04:00:00	132	127,287
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.25	19.2	:wt_pct	0.2668	2026-04-12T21:00:00	2026-04-13T13:00:00	17	32,533
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.24	19.2	:wt_pct	0.2296	2026-04-13T17:00:00	2026-04-13T23:00:00	7	27,996
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.23	19.2	:wt_pct	0.1672	2026-04-12T11:00:00	2026-04-12T16:00:00	6	20,386
Warning	:utilities	:pump_vibration_high	:PUMP_VIBRATION	6.8	6	:count	13.3	2026-03-26T04:00:00	2026-04-05T00:00:00	237	997.55
Info	:filtration	:vacuum_low	:FILTER_VACUUM	165.24	200	:mmwc	17.38	2026-04-17T06:00:00	2026-04-20T01:00:00	68	166,858
Info	:filtration	:vacuum_low	:FILTER_VACUUM	165.39	200	:mmwc	17.31	2026-02-09T06:00:00	2026-02-12T01:00:00	68	166,141
Info	:flotation	:reagent_demand_high	:REAGENT_FLOW	549.81	336	:l_ph	63.63	2026-04-20T13:00:00	2026-04-30T23:00:00	251	124,007
Info	:flotation	:reagent_demand_high	:REAGENT_FLOW	547.25	336	:l_ph	62.87	2026-01-01T05:00:00	2026-02-09T05:00:00	937	122,527
Info	:flotation	:reagent_demand_high	:REAGENT_FLOW	546.1	336	:l_ph	62.53	2026-02-12T14:00:00	2026-04-17T08:00:00	1,531	121,857

DID THE DETECTOR CATCH WHAT THE CAMPAIGN INJECTED?

Injected deviation	Role	Kind	Magnitude	Unit	Hours	Detected	Rule that caught it	Findings
:cooling_fouling	:reactor_temperature	:drift	6.2	:deg_c	264	0	:none	0
:sulfate_deficit	:sulfate_ratio	:bias	-1.7	:wt_pct	288	1	:sulfate_deficit	1
:grind_coarse	:grind_p80	:bias	42	:um	192	1	:grind_coarse	1
:reagent_starvation	:concentrate_grade	:bias	-1.5	:wt_pct	264	1	:concentrate_grade_low	3
:filter_cloth_blinding	:filtration_rate	:drift	-1.9	:t_per_m2_d	168	0	:none	0
:evaporator_fouling	:steam_flow	:drift	13.5	:t_ph	360	0	:none	0
:acid_strength_loss	:acid_strength	:drift	-1.6	:wt_pct	312	0	:none	0
:scrubber_breakthrough	:stack_fluoride	:bias	2.4	:count	384	1	:stack_fluoride_high	20
:mill_liner_wear	:mill_load	:drift	430	:kw	264	1	:mill_power_excess	1
:pump_cavitation	:pump_vibration	:bias	3.6	:count	240	1	:pump_vibration_high	1
:sparger_blockage	:product_n	:bias	-1.2	:wt_pct	264	1	:product_nitrogen_low	1
:crusher_gap_drift	:oversize_pct	:bias	6.5	:pct	264	0	:none	0
:rock_grade_drift	:rock_grade	:drift	1.3	:wt_pct	192	0	:none	0
:cyclone_roping	:slurry_density	:bias	6.4	:wt_pct	240	1	:slurry_density_high	1
:ammonia_overfeed	:product_n	:bias	1.1	:wt_pct	192	1	:product_nitrogen_high	3

The catalogue of the detector holds 20 rules, each with the signal it watches, the reference it compares against and the window it averages over; 'PHOSPHATE.RULE_CATALOGUE' prints them and the notebook shows them rule by rule.

Compliance register and corrective actions

● COMPLIANCE

88.1 % of clauses

42 clauses in 9 frameworks

● COMPLIANT

37 clauses

● AT RISK

0 clauses

within 10 % of the limit

● NON-COMPLIANT

5 clauses

outside the limit

THE SCORECARD, FRAMEWORK BY FRAMEWORK

Framework	Title		Clauses	Compliant	At risk	Non-compliant	Compliant %	
:eu_ied	EU Industrial Emissions Directive 2010/75		4	4	0	0	100	
:ghg_protocol	GHG Protocol corporate standard		4	2	0	2	50	
:ifa_bat	IFA/EFMA BAT for phosphate fertilisers		6	4	0	2	66.67	
:iso_14001	ISO 14001:2015 environmental management		5	5	0	0	100	
:iso_46001	ISO 46001:2019 water efficiency		3	3	0	0	100	
:iso_50001	ISO 50001:2018 energy management		6	5	0	1	83.33	
:iso_9001	ISO 9001:2015 quality management		8	8	0	0	100	
:phosphogypsum_code	Phosphogypsum stack management code		3	3	0	0	100	
:us_epa_40cfr63	US EPA 40 CFR 63 subpart AA		3	3	0	0	100	

Framework	Clause	Title	Metric	Value	Target	Unit	Margin %	Status
:iso_50001	:energy_review	Energy review of the process	:specific_energy	2.02	12.5	:gj_per_t	83.83	Compliant
:iso_50001	:enpi	Energy performance indicator	:spec_energy_vs_design	16.17	100	:pct	83.83	Compliant
:iso_50001	:baseline	Baseline data quality	:completeness	99.42	98	:pct	1.45	Compliant
:iso_50001	:objectives	Objectives and targets	:targets_met	85.71	70	:pct	22.45	Compliant
:iso_50001	:operational_control	Operational control	:critical_findings	31	0	:count	n/a	Noncompliant
:iso_50001	:monitoring	Monitoring and measurement	:metered_signals	69	40	:count	72.5	Compliant
:iso_14001	:aspects	Significant environmental aspects	:stack_fluoride	2.95	5	:count	40.91	Compliant
:iso_14001	:emissions	Emissions monitoring	:stack_dust	21.8	30	:count	27.34	Compliant
:iso_14001	:waste	Waste management	:gypsum_free_p2o5	0.4991	1	:wt_pct	50.09	Compliant
:iso_14001	:water	Water balance and discharge	:water_per_p2o5	3.36	8	:t_ph	58.01	Compliant
:iso_14001	:emergency	Emergency preparedness	:scrubber_ph	6.39	5.6	:count	14.14	Compliant
:iso_46001	:water_review	Water use review	:water_per_p2o5	3.36	8	:t_ph	58.01	Compliant
:iso_46001	:reuse	Water reuse	:water_recovery	70.3	65	:pct	8.15	Compliant
:iso_46001	:effluent	Effluent quality	:effluent_ph	7.4	8.5	:count	12.96	Compliant
:iso_9001	:product_spec	Product specification	:product_n	18.12	17.4	:wt_pct	4.13	Compliant
:iso_9001	:moisture	Product moisture	:product_moisture	1.75	2	:wt_pct	12.39	Compliant
:iso_9001	:solubility	Product solubility	:product_wsp	88.53	85	:wt_pct	4.15	Compliant
:iso_9001	:process_control	Control of production	:acid_strength	55.74	50.5	:wt_pct	10.38	Compliant
:iso_9001	:acid_specification	Merchant acid specification	:acid_so4	1.14	1.5	:wt_pct	23.7	Compliant
:iso_9001	:acid_fluorine	Fluorine of the merchant acid	:acid_f	0.3555	0.5	:wt_pct	28.89	Compliant
:iso_9001	:acid_solids	Solids of the merchant acid	:acid_solids	0.2953	0.5	:wt_pct	40.94	Compliant
:iso_9001	:acid_yield	Yield of the acid route	:acid_p2o5_yield	97.66	96.5	:pct	1.21	Compliant
:ghg_protocol	:scope_1	Scope 1 quantification	:scope_1_t	11,658	48	:t	-24,188	Noncompliant
:ghg_protocol	:scope_2	Scope 2 quantification	:scope_2_t	7,046.93	240	:t	-2,836.22	Noncompliant
:ghg_protocol	:intensity	Emissions intensity	:carbon_intensity	55.92	145	:kg_ph	61.43	Compliant
:ghg_protocol	:factors	Emission factor register	:factor_count	4	4	:count	0	Compliant
:eu_ied	:bat_emissions	BAT emission levels	:stack_fluoride	2.95	5	:count	40.91	Compliant
:eu_ied	:energy_efficiency	Energy efficiency of BAT	:specific_energy	2.02	12.5	:gj_per_t	83.83	Compliant

Framework	Clause	Title	Metric	Value	Target	Unit	Margin %	Status
:eu_ied	:monitoring	Continuous monitoring	:completeness	99.42	98	:pct	1.45	Compliant
:eu_ied	:waste	Waste management plan	:gypsum_free_p2o5	0.4991	1	:wt_pct	50.09	Compliant
:us_epa_40cfr63	:fluoride	Fluoride emission limit	:stack_fluoride	2.95	5	:count	40.91	Compliant
:us_epa_40cfr63	:scrubber	Scrubber operating limit	:scrubber_ph	6.39	5.6	:count	14.14	Compliant
:us_epa_40cfr63	:records	Record keeping	:completeness	99.42	98	:pct	1.45	Compliant
:ifa_bat	:recovery	P2O5 recovery of the route	:p2o5_recovery	82.83	94	:pct	-11.88	Noncompliant
:ifa_bat	:acid_demand	Specific acid demand	:acid_consumption	2.85	2.85	:t_ph	0.1048	Compliant
:ifa_bat	:acid_steam	Steam of the acid concentration	:acid_steam	0.4173	0.62	:t_ph	32.7	Compliant
:ifa_bat	:water_soluble	Water-soluble loss	:water_soluble_loss	2.34	2	:pct	-16.85	Noncompliant
:ifa_bat	:fluorine	Fluorine recovery	:fluorine_recovery	92	80	:pct	15	Compliant
:ifa_bat	:grind	Liberation of the ore	:grind_p80	152.96	170	:um	10.02	Compliant
:phosphogypsum_code	:stack_quality	Gypsum quality to the stack	:gypsum_free_p2o5	0.4991	1	:wt_pct	50.09	Compliant
:phosphogypsum_code	:water_return	Water return to the plant	:water_recovery	70.3	65	:pct	8.15	Compliant
:phosphogypsum_code	:dust	Dust control on the stack	:stack_dust	21.8	30	:count	27.34	Compliant

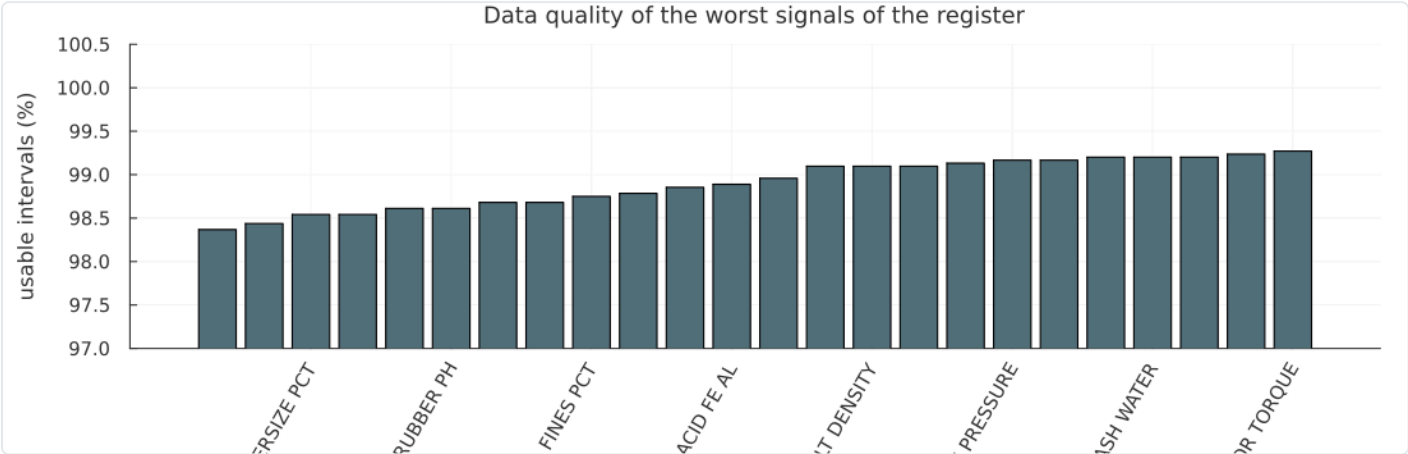
WHAT HAS TO BE DONE ABOUT IT

Framework	Clause	Metric	Value	Target	Gap %	Status	Corrective action
:ghg_protocol	:scope_1	:scope_1_t	11,658	48	-24,188	Noncompliant	Review the operating window with the process engineer
:ghg_protocol	:scope_2	:scope_2_t	7,046.93	240	-2,836.22	Noncompliant	Review the operating window with the process engineer
:ifa_bat	:water_soluble	:water_soluble_loss	2.34	2	-16.85	Noncompliant	Review the operating window with the process engineer
:ifa_bat	:recovery	:p2o5_recovery	82.83	94	-11.88	Noncompliant	Review the reagent dose and the grind of the blend
:iso_50001	:operational_control	:critical_findings	31	0	n/a	Noncompliant	Close the critical findings of the diagnostics register

Data quality, reconciliation and the instrument register

QUALITY OF THE HISTORIAN

Quantity	Value
Intervals	2,880 hourly intervals of 69 signals
Operating intervals	2,736
Completeness	99.42 % of the register
Defect runs	140 runs over 1146 intervals
Causes	:faulty 567, :measured 196944, :missing 579, :substituted 630
Status	:compliant



Completeness of the historian, the worst signals of the instrument register

EVERY RUN OF UNUSABLE DATA OF THE CAMPAIGN

Signal	From	To	Intervals	Cause
:ACID_COLOUR	2026-01-01T23:00:00	2026-01-02T08:00:00	10	Faulty
:OVERSIZE_PCT	2026-01-02T06:00:00	2026-01-02T19:00:00	14	Missing
:OVERSIZE_PCT	2026-01-04T07:00:00	2026-01-04T14:00:00	8	Missing
:CYCLONE_P80	2026-01-06T10:00:00	2026-01-06T21:00:00	12	Missing
:ACID_SO4	2026-01-08T07:00:00	2026-01-08T19:00:00	13	Faulty
:AMMONIA_FLOW	2026-01-08T14:00:00	2026-01-08T17:00:00	4	Missing
:PROCESS_WATER_FLOW	2026-01-09T03:00:00	2026-01-09T08:00:00	6	Missing
:F_WASH_WATER	2026-01-10T23:00:00	2026-01-11T01:00:00	3	Missing
:PRODUCT_WSP	2026-01-11T18:00:00	2026-01-12T01:00:00	8	Faulty
:PRODUCT_MOISTURE	2026-01-13T20:00:00	2026-01-14T07:00:00	12	Faulty
:DRYER_OUTLET_TEMP	2026-01-13T21:00:00	2026-01-14T03:00:00	7	Missing
:FINES_PCT	2026-01-13T23:00:00	2026-01-14T01:00:00	3	Faulty
:SLURRY_SG	2026-01-15T23:00:00	2026-01-16T06:00:00	8	Missing
:STACK_DUST	2026-01-16T06:00:00	2026-01-16T11:00:00	6	Faulty
:STACK_DUST	2026-01-17T06:00:00	2026-01-17T18:00:00	13	Faulty
:TAIL_P2O5	2026-01-18T09:00:00	2026-01-18T14:00:00	6	Missing
:OVERSIZE_PCT	2026-01-18T20:00:00	2026-01-19T05:00:00	10	Faulty
:DRYER_OUTLET_TEMP	2026-01-19T09:00:00	2026-01-19T15:00:00	7	Faulty
:RECYCLE_RATIO	2026-01-19T14:00:00	2026-01-19T19:00:00	6	Faulty
:COOLER_OUTLET_TEMP	2026-01-20T12:00:00	2026-01-20T15:00:00	4	Missing
:COOLING_SUPPLY_T	2026-01-22T04:00:00	2026-01-22T07:00:00	4	Faulty
:OVERSIZE_PCT	2026-01-22T04:00:00	2026-01-22T10:00:00	7	Faulty

Signal	From	To	Intervals	Cause
:SULPHUR_FEED	2026-01-22T23:00:00	2026-01-23T09:00:00	11	Missing
:PRODUCT_WSP	2026-01-23T16:00:00	2026-01-24T04:00:00	13	Faulty
:COOLER_OUTLET_TEMP	2026-01-23T17:00:00	2026-01-23T21:00:00	5	Missing
:ACID_FE_AL	2026-01-24T12:00:00	2026-01-25T00:00:00	13	Missing
:CONVERTER_TEMP	2026-01-24T20:00:00	2026-01-25T04:00:00	9	Missing
:OVERSIZE_PCT	2026-01-24T23:00:00	2026-01-25T01:00:00	3	Faulty
:ROCK_FEED	2026-01-26T08:00:00	2026-01-26T21:00:00	14	Faulty
:ACID_SOLIDS	2026-01-26T14:00:00	2026-01-26T19:00:00	6	Faulty
:PROCESS_WATER_FLOW	2026-01-26T22:00:00	2026-01-27T07:00:00	10	Faulty
:SCRUBBER_PH	2026-01-27T09:00:00	2026-01-27T16:00:00	8	Missing
:MILL_SOUND_DB	2026-01-31T02:00:00	2026-01-31T08:00:00	7	Faulty
:DRYER_OUTLET_TEMP	2026-01-31T04:00:00	2026-01-31T15:00:00	12	Faulty
:CONVERTER_TEMP	2026-01-31T18:00:00	2026-02-01T07:00:00	14	Missing
:STRONG_ACID_P2O5	2026-02-03T11:00:00	2026-02-03T13:00:00	3	Faulty
:CYCLONE_PRESSURE	2026-02-03T22:00:00	2026-02-04T07:00:00	10	Missing
:CONC_P2O5	2026-02-04T08:00:00	2026-02-04T10:00:00	3	Missing
:MILL_POWER	2026-02-05T01:00:00	2026-02-05T04:00:00	4	Missing
:TAIL_P2O5	2026-02-05T15:00:00	2026-02-06T02:00:00	12	Faulty
:GRANULATOR_TORQUE	2026-02-07T09:00:00	2026-02-07T11:00:00	3	Missing
:FILTER_VACUUM	2026-02-09T06:00:00	2026-02-09T10:00:00	5	Faulty
:WASH_WATER_FLOW	2026-02-09T07:00:00	2026-02-09T14:00:00	8	Missing
:ACID_F	2026-02-09T23:00:00	2026-02-10T01:00:00	3	Faulty
:ACID_FE_AL	2026-02-10T08:00:00	2026-02-10T15:00:00	8	Missing
:ACID_F	2026-02-10T16:00:00	2026-02-11T04:00:00	13	Missing
:CYCLONE_P80	2026-02-11T01:00:00	2026-02-11T07:00:00	7	Missing
:PRODUCT_FLOW	2026-02-12T01:00:00	2026-02-12T09:00:00	9	Faulty
:CYCLONE_P80	2026-02-13T02:00:00	2026-02-13T10:00:00	9	Faulty
:SCRUBBER_PH	2026-02-14T21:00:00	2026-02-15T08:00:00	12	Faulty
:CONVERTER_TEMP	2026-02-15T08:00:00	2026-02-15T17:00:00	10	Missing
:SCRUBBER_PH	2026-02-15T23:00:00	2026-02-16T12:00:00	14	Missing
:WEAK_ACID_P2O5	2026-02-16T05:00:00	2026-02-16T14:00:00	10	Faulty
:CRUSHER_POWER	2026-02-17T09:00:00	2026-02-17T17:00:00	9	Faulty
:GYPSUM_FLOW	2026-02-19T19:00:00	2026-02-19T21:00:00	3	Faulty
:TURBINE_POWER	2026-02-20T00:00:00	2026-02-20T06:00:00	7	Missing
:FINES_PCT	2026-02-20T03:00:00	2026-02-20T06:00:00	4	Missing
:GRAN_BED_TEMP	2026-02-20T22:00:00	2026-02-21T00:00:00	3	Faulty
:TURBINE_POWER	2026-02-20T22:00:00	2026-02-21T08:00:00	11	Missing
:FLOT_FEED_P2O5	2026-02-21T07:00:00	2026-02-21T11:00:00	5	Faulty
:GYPSUM_FREE_P2O5	2026-02-22T18:00:00	2026-02-23T07:00:00	14	Missing
:STEAM_FLOW	2026-02-25T01:00:00	2026-02-25T03:00:00	3	Missing
:GYPSUM_FLOW	2026-02-26T07:00:00	2026-02-26T11:00:00	5	Faulty
:MELT_DENSITY	2026-02-26T20:00:00	2026-02-27T07:00:00	12	Faulty
:CYCLONE_P80	2026-02-27T01:00:00	2026-02-27T14:00:00	14	Missing
:MELT_DENSITY	2026-02-28T23:00:00	2026-03-01T02:00:00	4	Missing
:EVAP_FEED_FLOW	2026-03-01T01:00:00	2026-03-01T03:00:00	3	Faulty

Signal	From	To	Intervals	Cause
:RECYCLE_ACID_FLOW	2026-03-01T23:00:00	2026-03-02T09:00:00	11	Faulty
:GRANULATOR_TORQUE	2026-03-03T10:00:00	2026-03-03T13:00:00	4	Missing
:F_WASH_WATER	2026-03-04T12:00:00	2026-03-04T21:00:00	10	Missing
:SULPHUR_FEED	2026-03-04T16:00:00	2026-03-05T04:00:00	13	Missing
:ACID_COLOUR	2026-03-06T04:00:00	2026-03-06T14:00:00	11	Missing
:ROM_FEED	2026-03-06T08:00:00	2026-03-06T20:00:00	13	Faulty
:FILTER_VACUUM	2026-03-06T18:00:00	2026-03-07T01:00:00	8	Faulty
:CYCLONE_PRESSURE	2026-03-06T20:00:00	2026-03-07T04:00:00	9	Faulty
:WASH_WATER_FLOW	2026-03-07T03:00:00	2026-03-07T13:00:00	11	Faulty
:MERCHANT_ACID_FLOW	2026-03-08T13:00:00	2026-03-09T02:00:00	14	Faulty
:CONC_P2O5	2026-03-10T03:00:00	2026-03-10T15:00:00	13	Missing
:RECYCLE_RATIO	2026-03-10T13:00:00	2026-03-10T20:00:00	8	Missing
:PRODUCT_FLOW	2026-03-11T05:00:00	2026-03-11T08:00:00	4	Faulty
:STRONG_ACID_P2O5	2026-03-11T05:00:00	2026-03-11T11:00:00	7	Missing
:F_WASH_WATER	2026-03-11T06:00:00	2026-03-11T15:00:00	10	Missing
:GRANULATOR_TORQUE	2026-03-11T18:00:00	2026-03-12T00:00:00	7	Missing
:PRODUCT_MOISTURE	2026-03-11T19:00:00	2026-03-12T05:00:00	11	Missing
:ROCK_P2O5	2026-03-12T18:00:00	2026-03-13T00:00:00	7	Faulty
:SLURRY_SG	2026-03-13T19:00:00	2026-03-13T21:00:00	3	Missing
:EVAP_VACUUM	2026-03-14T00:00:00	2026-03-14T12:00:00	13	Faulty
:PRODUCT_FLOW	2026-03-17T23:00:00	2026-03-18T10:00:00	12	Missing
:CYCLONE_P80	2026-03-18T10:00:00	2026-03-18T12:00:00	3	Faulty
:ROCK_FEED	2026-03-20T01:00:00	2026-03-20T03:00:00	3	Faulty
:RECYCLE_RATIO	2026-03-22T16:00:00	2026-03-22T21:00:00	6	Faulty
:FREE_SO4	2026-03-22T17:00:00	2026-03-22T22:00:00	6	Missing
:PRODUCT_FLOW	2026-03-23T14:00:00	2026-03-23T19:00:00	6	Missing
:COOLING_RETURN_T	2026-03-24T00:00:00	2026-03-24T09:00:00	10	Missing
:FLOT_FEED_P2O5	2026-03-24T15:00:00	2026-03-25T04:00:00	14	Faulty
:FLOT_FEED_P2O5	2026-03-25T08:00:00	2026-03-25T15:00:00	8	Faulty
:SCRUBBER_PH	2026-03-25T12:00:00	2026-03-25T17:00:00	6	Faulty
:STEAM_FLOW	2026-03-25T17:00:00	2026-03-25T21:00:00	5	Missing
:H2SO4_FLOW	2026-03-25T18:00:00	2026-03-26T06:00:00	13	Missing
:MELT_DENSITY	2026-03-25T19:00:00	2026-03-26T04:00:00	10	Missing
:STEAM_FLOW	2026-03-26T02:00:00	2026-03-26T03:00:00	2	Missing
:GRAN_BED_TEMP	2026-03-26T23:00:00	2026-03-27T01:00:00	3	Missing
:FLOT_FEED_P2O5	2026-03-28T02:00:00	2026-03-28T14:00:00	13	Missing
:WASH_WATER_FLOW	2026-03-29T07:00:00	2026-03-29T13:00:00	7	Faulty
:EVAP_DENSITY	2026-04-01T02:00:00	2026-04-01T05:00:00	4	Faulty
:CYCLONE_PRESSURE	2026-04-01T10:00:00	2026-04-01T14:00:00	5	Missing
:FILTER_RATE	2026-04-01T10:00:00	2026-04-01T17:00:00	8	Faulty
:MERCHANT_ACID_FLOW	2026-04-02T12:00:00	2026-04-02T23:00:00	12	Faulty
:ROCK_P2O5	2026-04-04T14:00:00	2026-04-05T03:00:00	14	Missing
:FINES_PCT	2026-04-06T04:00:00	2026-04-06T11:00:00	8	Faulty
:WASH_WATER_FLOW	2026-04-06T05:00:00	2026-04-06T13:00:00	9	Faulty
:GRANULATOR_TORQUE	2026-04-06T20:00:00	2026-04-07T02:00:00	7	Faulty

Signal	From	To	Intervals	Cause
:FILTER_VACUUM	2026-04-07T13:00:00	2026-04-07T18:00:00	6	Faulty
:ACID_COLOUR	2026-04-10T19:00:00	2026-04-10T22:00:00	4	Faulty
:FILTER_VACUUM	2026-04-12T13:00:00	2026-04-12T15:00:00	3	Missing
:PRODUCT_WSP	2026-04-14T06:00:00	2026-04-14T18:00:00	13	Faulty
:ACID_SOLIDS	2026-04-14T17:00:00	2026-04-15T06:00:00	14	Missing
:SO2_CONC	2026-04-15T03:00:00	2026-04-15T11:00:00	9	Missing
:FINES_PCT	2026-04-15T08:00:00	2026-04-15T15:00:00	8	Missing
:PRODUCT_WSP	2026-04-16T02:00:00	2026-04-16T09:00:00	8	Missing
:FINES_PCT	2026-04-16T03:00:00	2026-04-16T15:00:00	13	Faulty
:CRUSHER_POWER	2026-04-17T01:00:00	2026-04-17T07:00:00	7	Missing
:SULPHUR_FEED	2026-04-17T10:00:00	2026-04-17T23:00:00	14	Missing
:MILL_FEED_WATER	2026-04-17T11:00:00	2026-04-17T16:00:00	6	Missing
:MILL_FEED_WATER	2026-04-17T19:00:00	2026-04-17T21:00:00	3	Missing
:FILTER_RATE	2026-04-17T20:00:00	2026-04-17T23:00:00	4	Missing
:FILTER_RATE	2026-04-18T00:00:00	2026-04-18T10:00:00	11	Faulty
:GRAN_BED_TEMP	2026-04-18T02:00:00	2026-04-18T05:00:00	4	Missing
:PRODUCT_FLOW	2026-04-19T13:00:00	2026-04-19T19:00:00	7	Faulty
:GYPSUM_FLOW	2026-04-22T01:00:00	2026-04-22T08:00:00	8	Faulty
:RECYCLE_ACID_FLOW	2026-04-22T05:00:00	2026-04-22T13:00:00	9	Faulty
:ROCK_FEED	2026-04-24T09:00:00	2026-04-24T12:00:00	4	Faulty
:GRAN_BED_TEMP	2026-04-24T10:00:00	2026-04-24T23:00:00	14	Missing
:TURBINE_POWER	2026-04-26T04:00:00	2026-04-26T11:00:00	8	Missing
:ROCK_FEED	2026-04-27T04:00:00	2026-04-27T17:00:00	14	Faulty
:ACID_FE_AL	2026-04-27T05:00:00	2026-04-27T15:00:00	11	Faulty
:WASH_WATER_FLOW	2026-04-27T23:00:00	2026-04-28T10:00:00	12	Missing
:GYPSUM_FREE_P2O5	2026-04-28T05:00:00	2026-04-28T09:00:00	5	Faulty
:MERCHANT_ACID_FLOW	2026-04-28T18:00:00	2026-04-28T21:00:00	4	Faulty
:EVAP_DENSITY	2026-04-29T23:00:00	2026-04-30T04:00:00	6	Faulty

The flows of the campaign were reconciled against the stoichiometry of the reaction section: the weighted least squares adjustment leaves a global test statistic of 21,788 against a critical value of 11.07 for 5 balances, so the measurements are inconsistent with the balances.

Signal	Measured	Reconciled	Adjustment	In sigma	Unit	Gross error
:P2O5_FEED_RATE	122.25	38.83	-83.42	-104.27	:t_ph	1
:H2SO4_FLOW	348.05	380.29	32.24	10.75	:t_ph	1
:SULPHUR_FEED	113.62	219.25	105.63	88.03	:t_ph	1
:GYPSUM_FLOW	572.42	770.42	198.01	49.5	:t_ph	1
:AMMONIA_FLOW	44.39	58.76	14.36	17.95	:t_ph	1
:PRODUCT_FLOW	199.01	265.85	66.84	16.71	:t_ph	1

Method, units and assumptions

Area	Streams	Equipment	Unit operations	Instrument tags
Beneficiation		8	9	4
Flotation		9	5	4
Attack		5	3	3
Filtration		6	5	4
Evaporation		3	3	3
SulphuricAcidPlant		6	7	4
Granulation		15	8	8
Utilities		8	3	2
Tailings		1	1	2

- The **design** of the plant is registered in `process.jl`: capacities, targets, the flowsheet and the instrument register. Every model of the package reads it, so a capacity change propagates to the twin, the optimiser, the controller and the report.
- The **campaign** is a seeded grey-box model of that design (`plantdata.jl`): the production schedule, the ore blend, the weather and the injected deviations are all functions of the seed, so every figure and every PDF can be reproduced from the seed printed on the cover page.
- The **stoichiometry** is computed from the molar masses in `measures.jl`: 2.3 t of H2SO4 and 4.04 t of gypsum per tonne of P2O5 in fluorapatite, plus the demand of the free carbonate of the ore. The assays of the ore bodies are representative published figures, not the assay of a particular mine.
- The **twin** is written in ModelingToolkit: a square nonlinear balance of 15 unknowns for the steady state, and 11 differential equations on an hourly clock for the dynamics. Both report the residual of every equation they solve.
- The **controller** is a linear MPC over the linearised twin with a Kalman filter, closed on the nonlinear twin as the plant; the baseline is a PI controller tuned on the same steady-state gains, so the comparison is like for like.
- The **soft sensors** are linear models of the variables that are measured slowly or not at all, validated on a holdout and equipped with an applicability domain: outside the conditions they were fitted on, they report that they cannot answer.
- The **cost and carbon** of the site use the register of `process.jl` (USD) and the factors of the GHG Protocol: 0.371 kgCO2e/kWh for the grid and 50.29 kgCO2e/GJ for the fuel.

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